

IoT Communications

The connections among IoT objects and between them and the global Internet are enabled by different technologies:

- Wired (USB, Ethernet, Powerline)
- **Wireless (Wifi, Bluetooth, ZigBee, IEEE 802.15.x, LoRA, ...)**

Communications take place according to multiple network and association models:

- Infrastructured or ad-hoc
 - Basic Service Set, Independent Service Set
- Single-hop or multi-hop
 - Sink nodes, cluster heads...
- Point-to-Point or Point-to-Multi Point

Routing is a fundamental issue

- Mobile Routing
- Energy efficient routing

The IoT Protocol Stack (reminder...)

HTTP, DNS, XMPP, MQTT, CoAp, TLS, SSL	Application layer
TCP, UDP	Transport Layer
IP(V4, V6), 6LowPAN	Internet Layer
Ethernet, 802.11 WiFi, 802.15.4	Link Layer
Electric, optical or RF transmission	Physical Layer

We are essentially interested in Radio Frequency-based communications, that is the “*silver bullet*” technology that enabled the growth and popularity of IoT due to:

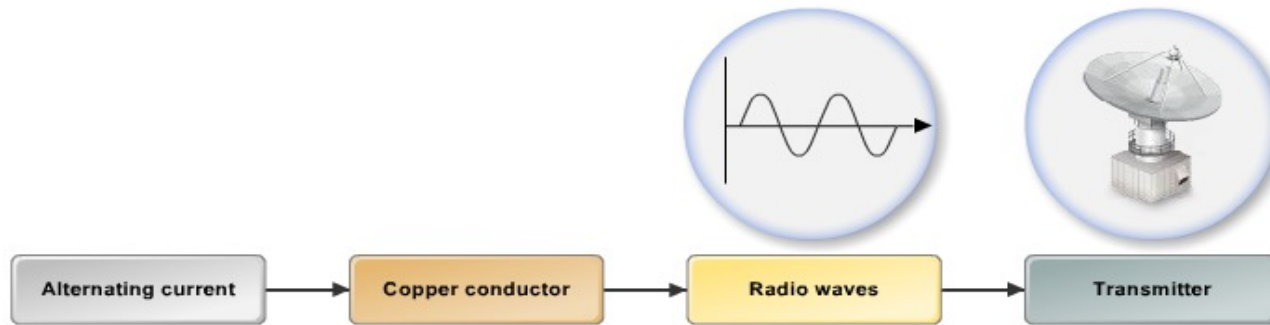
- Ubiquitous and pervasive presence
- Flexibility and Ease of installation
- Extremely limited costs

The Physical Layer: radio waves

- Wireless Communications: Any communication system that uses a naturally occurring communication channel, such as the air.
 - Communications channel is natural (air)
 - poor quality: fading, shadowing, weather, etc.
 - Medium regulated by governments
 - frequency allocation, licensing, etc.
- Examples:
 - Sonar
 - Broadcast: Radio, TV, pagers, satellite TV, etc.
 - Two Way: walkie talkie, cell phones, satellite phones,
 - Wireless Local Area Networks, etc.

Radio waves generation

- When an alternating current (AC) passes through a copper conductor it generates radio waves (*i.e.*, an *electro-magnetic field*) which are propagated into the air using an *antenna*
- In the same way, if a copper conductor is affected by an electro-magnetic fields (radio waves) some currents/electrical voltages varying over time are generated



- radio waves travel at a velocity of 186.000 miles per second (lower than the speed of light c)
- radio waves become weaker as they travel a long distance

Radio waves generation

- Maxwell's equations describe how electric and magnetic fields are generated by currents

$$\nabla \cdot \mathbf{D} = \rho$$

Gauss' Law for Electricity

describes the relationship between an electric field and electric charges

$$\nabla \cdot \mathbf{B} = 0$$

Gauss' Law for Magnetism

states that electric charges have no magnetic analogues and the net outflow of the magnetic field through a closed surface is zero

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Faraday's Law of Induction

describes how a time-varying magnetic field corresponds to curl of an electric field (a rotating magnet creates a changing magnetic field and an electric field in a nearby wire).

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Ampere's Law

magnetic fields relate to electric current and to changing electric fields

where:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}$$

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \quad \text{in the general case}$$

$\mathbf{E}$ = Electric field

ε_0 = permittivity

μ_0 = permeability

$\mathbf{P}$ = Polarization

ρ = charge density

$\mathbf{J}$ = current density

$\mathbf{H}$ = Magnetic field strength

$\nabla \cdot$ = divergence

$\mathbf{B}$ = Magnetic field

$\mathbf{D}$ = Electric displacement

$\mathbf{M}$ = Magnetization

$\nabla \times$ = curl (rotation)

Understanding Maxwell equations in differential form

- We showed a system of differential equations on vectorial elements. Understanding the notation used can be useful:

- The gradient operator can be defined as: $\nabla \equiv \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix}$

- The divergence is the scalar product of a gradient and a vector $\mathbf{F}$:

$$\nabla \cdot \mathbf{F} = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \cdot \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \frac{\partial}{\partial x} F_x + \frac{\partial}{\partial y} F_y + \frac{\partial}{\partial z} F_z.$$

- The rotation is the vector product of the gradient and $\mathbf{F}$:

$$\nabla \times \mathbf{F} = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \times \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \partial F_z/\partial y - \partial F_y/\partial z \\ \partial F_x/\partial z - \partial F_z/\partial x \\ \partial F_y/\partial x - \partial F_x/\partial y \end{bmatrix}.$$

Understanding Maxwell equations in differential form

- Thus, for example: $\nabla \cdot \mathbf{B} = 0$

means:
$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0$$

- Analogously: $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

is expandable into:
$$\begin{cases} \frac{\partial B_x}{\partial t} = \frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y}, \\ \frac{\partial B_y}{\partial t} = \frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z}, \\ \frac{\partial B_z}{\partial t} = \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}, \end{cases}$$

The wave equation

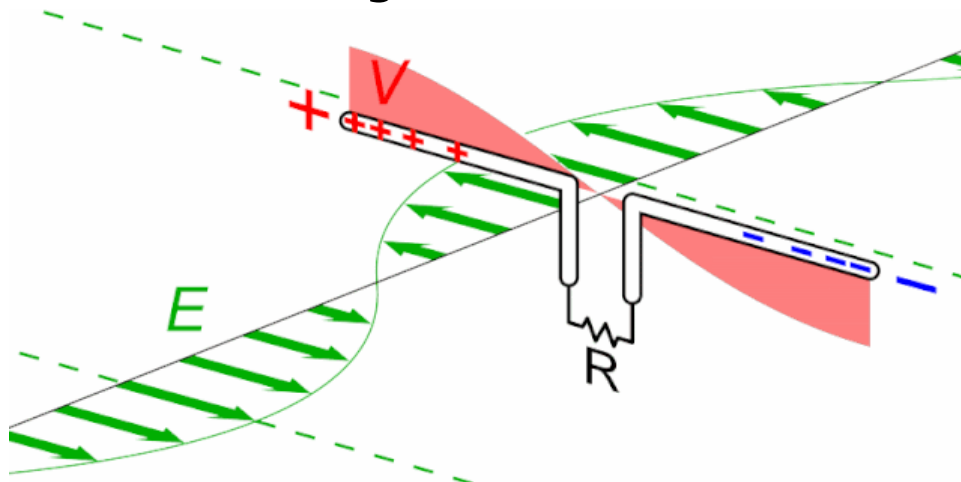
- The wave equations derive directly from the Maxwell equations (by substituting fields $\mathbf{D}$ and $\mathbf{B}$ and combining the resulting equations):

$$\begin{aligned}\nabla \times \nabla \times \mathbf{E} + \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} &= -\mu_0 \frac{\partial}{\partial t} \left(\mathbf{J} + \frac{\partial \mathbf{P}}{\partial t} + \nabla \times \mathbf{M} \right) \\ \nabla \times \nabla \times \mathbf{H} + \frac{1}{c^2} \frac{\partial^2 \mathbf{H}}{\partial t^2} &= \nabla \times \mathbf{J} + \nabla \times \frac{\partial \mathbf{P}}{\partial t} - \frac{1}{c^2} \frac{\partial^2 \mathbf{M}}{\partial t^2}\end{aligned}$$

- It gives the radiated electric field E from a time-varying current $\mathbf{J}$ where in the round brackets we can see the total current density.
- This is the foundation of antenna theory since it reveals how the antenna is related to radio waves
- It proves that only a time-varying current can generate a radio wave – not static charges or DC current
- Antenna design is basically how to control the current distribution $\mathbf{J}$ and hence to obtain the desired radiated field E .

Radio waves generation

- A variable electric field generates a variable magnetic field (generalized Ampère's theorem), which in turn generates a variable electric field (Faraday-Neumann-Lenz law) and so on
- The oscillation of an electric field, i.e. the variation of its intensity in a point as time varies, generates the oscillation of a magnetic field in nearby points...
- ...which in turn generates the oscillation of an electric field in nearby points and so on: the oscillation propagates in space in the form of an electromagnetic wave

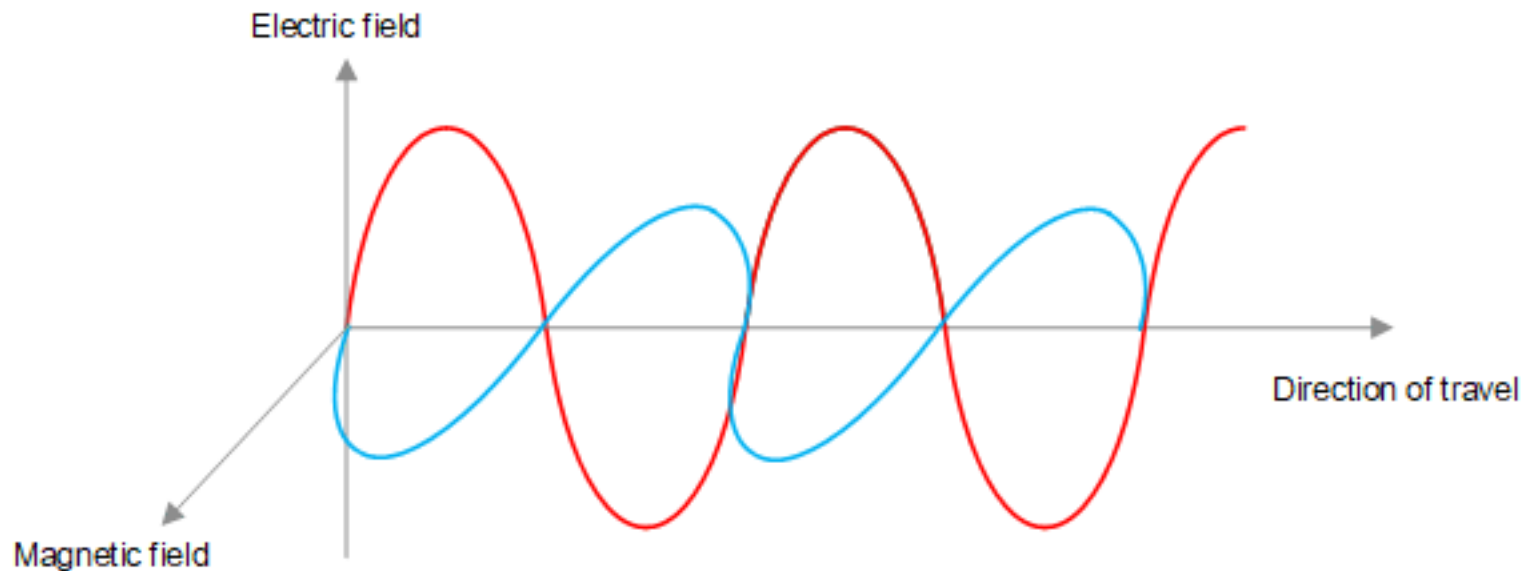


Radio waves generation

- No Matter What... its Just a wave

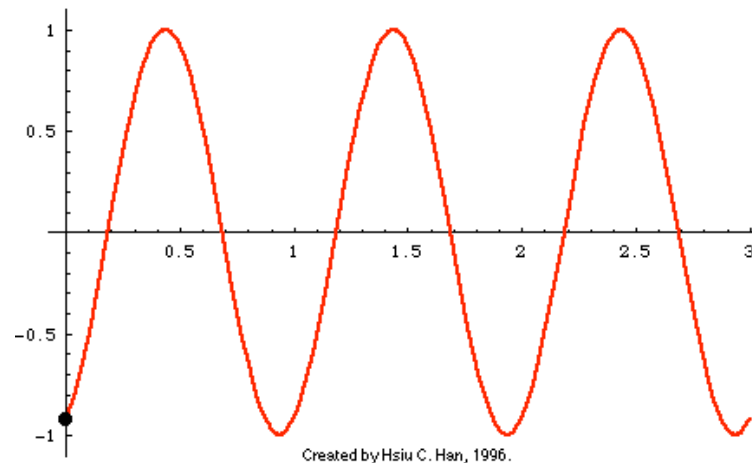
$$E(t) = A(t) \cos[\omega t + \phi(t)]$$

- $A(t)$: amplitude of the wave
- ω : radian frequency of the wave
- $\phi(t)$: phase of the wave



Radio Frequency

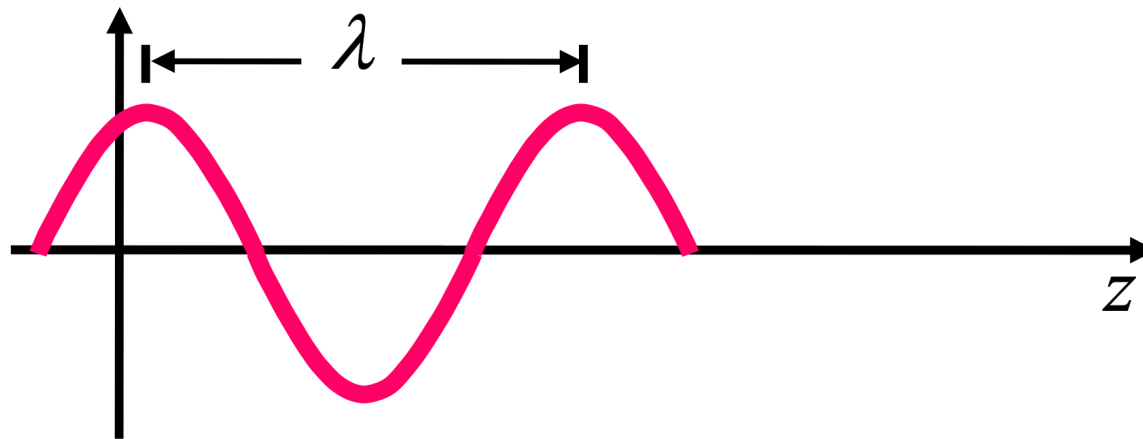
- *Waves refer to* cyclical, oscillatory processes, where frequency is defined as the number of cycles per time unit.



- The relation between the frequency and the period T of a repeating event or oscillation is given by $f = 1/T$
- the full range of all frequencies of electromagnetic radiation is defined as *electromagnetic spectrum*

Wavelength and Velocity

- How fast is the wave traveling?
- Velocity is a reference distance divided by a reference time



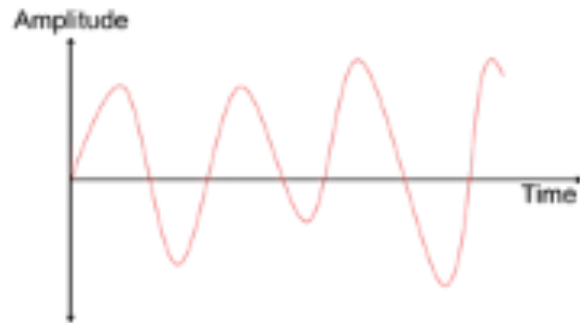
The wave moves one wavelength, λ , in one period, τ .

- The velocity is the wavelength / period:

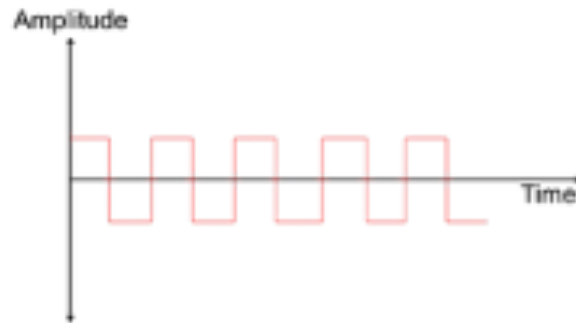
$$c = \lambda / \tau = \lambda f$$

Signals

- We will use radio waves to transport signals
- A signal is a physical representation of data depending on time and location
- Classification of signals:
 - analog signals = continuous time and continuous values
 - **digital signals = discrete time and discrete values**



Analog Signal



Digital Signal

Periodic Signals

- A periodic signal is characterized by:
 - period T , frequency f , amplitude A , phase shift Φ
 - A sine wave can be seen as special periodic signal for a carrier:

$$g(t) = A \sin(2\pi ft + \Phi)$$

- A periodic signal includes many frequencies
 - According to Fourier theorem it can be seen as a combination of multiple sinusoids whose frequencies are integer multiples of the fundamental frequency $f = 1/T$

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft)$$

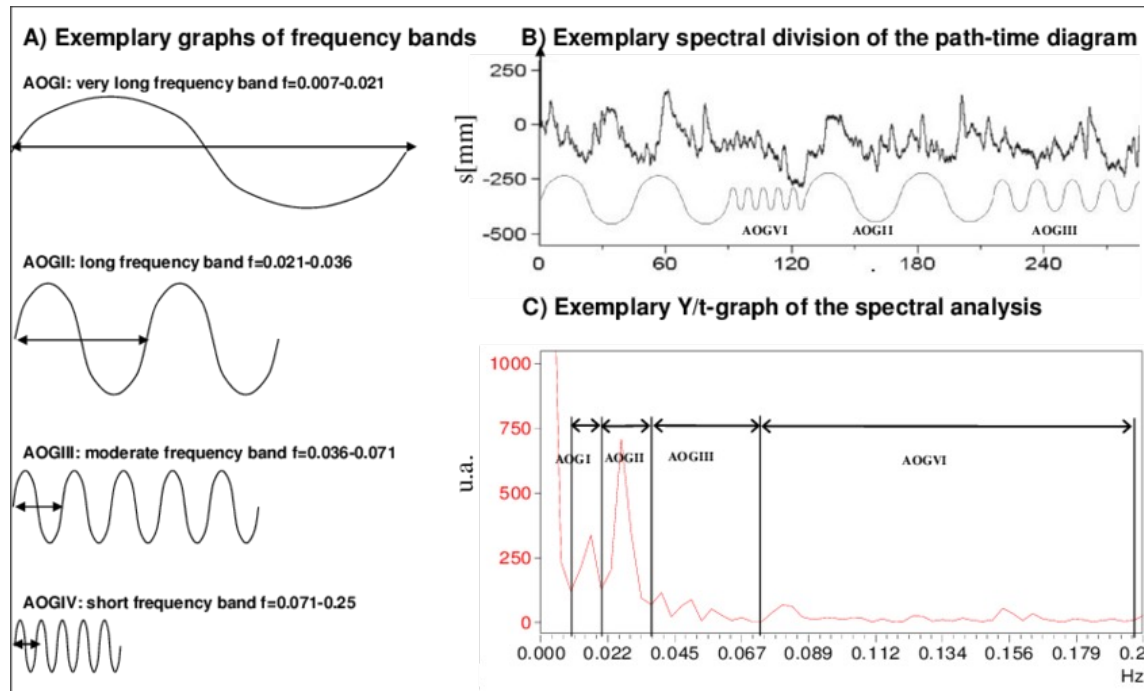
and whose amplitudes can be calculated as:

$$a_n = \frac{2}{T} \int_0^T g(t) \sin(2\pi nft) dt \quad b_n = \frac{2}{T} \int_0^T g(t) \cos(2\pi nft) dt \quad c = \frac{2}{T} \int_0^T g(t) dt$$

- The spectrum of a signal is the range of frequencies in it
- The bandwidth is the width (extension) of the spectrum

Frequency Carriers/Channels

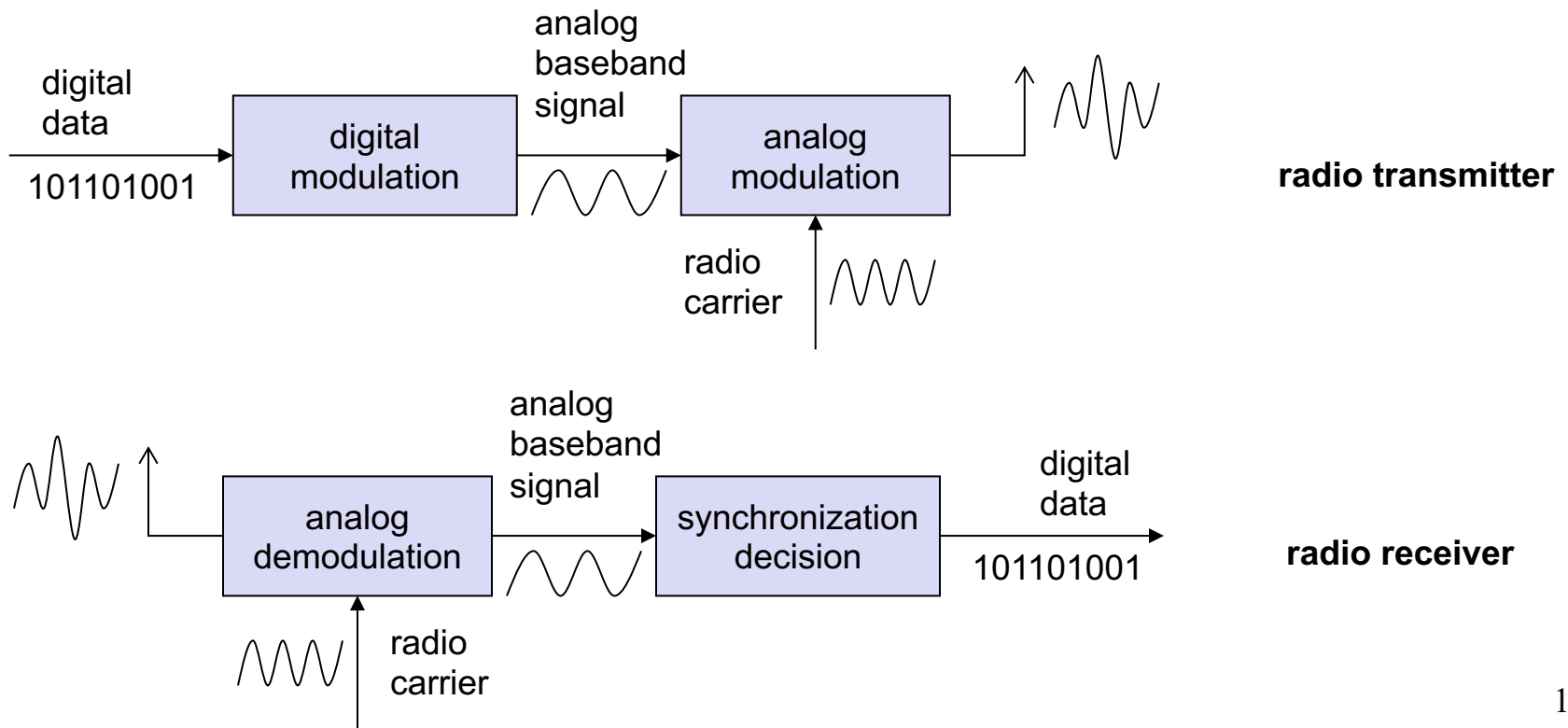
- Information is carried over a well-defined frequency band (channel) by using a periodic signal (carrier) whose properties vary over time.
- Each channel has a fixed band (in KHz) and Capacity (bit-rate)
- Different frequency bands (channels) can be used to transmit information in parallel and independently.



Modulation and demodulation

—Digital modulation

- digital data is translated into an analog signal to be transmitted on a specific frequency/channel
- ASK, FSK, PSK
 - differences in spectral efficiency, power efficiency, robustness



Modulation and demodulation

- Modulation is the process of varying one or more properties of a periodic waveform, called the carrier signal, with a separate signal that typically contains information to be transmitted.
- The purpose is to impress the information on the carrier wave

Signal characteristics that can be modified

$$\text{signal } x(t) = A \cos(2\pi ft + \Phi)$$

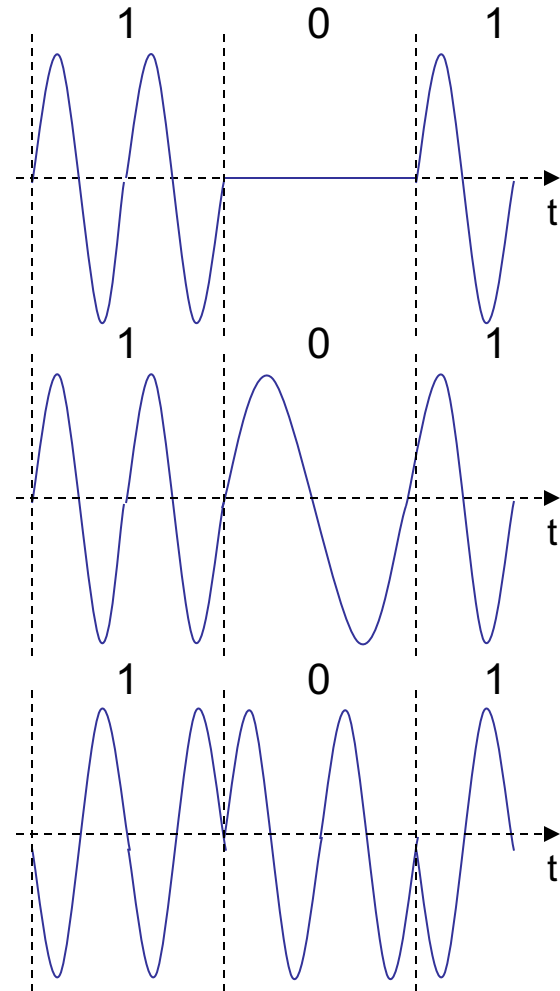
A – amplitude

f – frequency

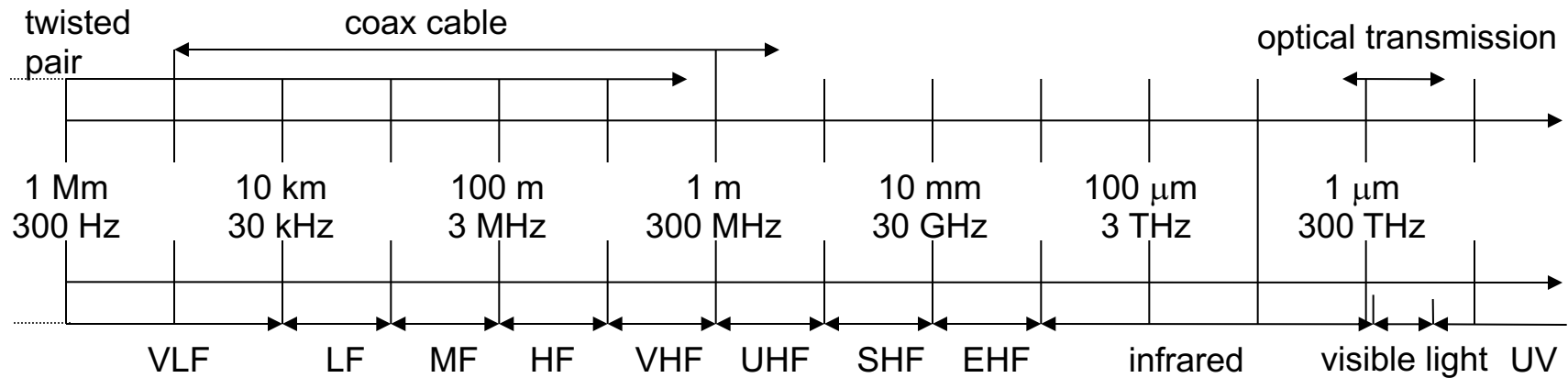
Φ – phase (initial angle of the sinusoidal function at its origin)

Digital modulation

- Known as Shift Keying
- Amplitude Shift Keying (ASK):
 - very simple
 - low bandwidth requirements
 - very susceptible to interference
- Frequency Shift Keying (FSK):
 - needs larger bandwidth
- Phase Shift Keying (PSK):
 - more complex
 - robust against interference
- Many advanced variants



Frequencies for communication



- VLF = Very Low Frequency
- LF = Low Frequency
- MF = Medium Frequency
- HF = High Frequency
- VHF = Very High Frequency

- UHF = Ultra High Frequency
- SHF = Super High Frequency
- EHF = Extra High Frequency
- UV = Ultraviolet Light

- Frequency and wavelength: $\lambda = c/f$
- wavelength λ , speed of light $c \cong 3 \times 10^8 \text{ m/s}$, frequency f

Wireless frequency allocation

- Radio frequencies range from 9KHz to 400GHZ (ITU)
- Microwave frequency range
 - 1 GHz to 40 GHz
 - Directional beams possible
 - Suitable for point-to-point transmission
 - Used for satellite communications
- Radio frequency range
 - 30 MHz to 1 GHz
 - Suitable for omnidirectional applications
- Infrared frequency range
 - Roughly, 3×10^{11} to 2×10^{14} Hz
 - Useful in local point-to-point multipoint applications within confined areas

Frequencies for mobile communication

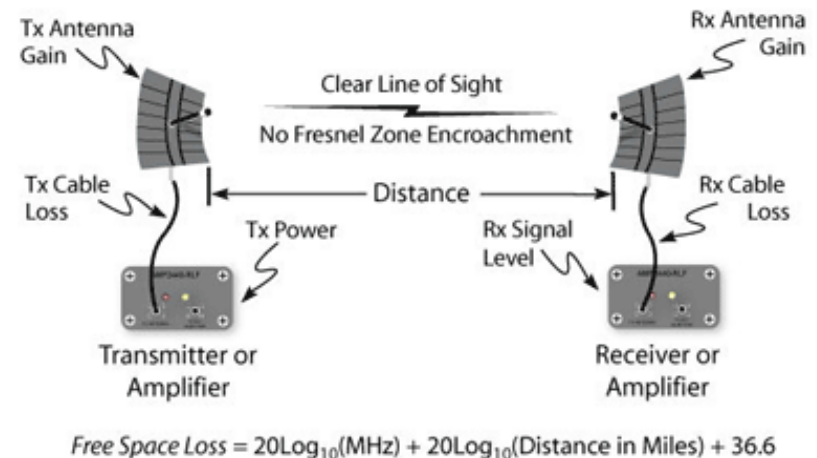
- VHF-/UHF-ranges for mobile radio
 - simple, small antenna for cars
 - deterministic propagation characteristics, reliable connections
- SHF and higher for directed radio links, satellite communication
 - small antenna, focusing
 - large bandwidth available
- Wireless LANs use frequencies in UHF to SHF spectrum
 - some systems planned up to EHF
 - limitations due to absorption by water and oxygen molecules (resonance frequencies)
 - weather dependent fading, signal loss caused by heavy rainfall etc.

Frequency regulations

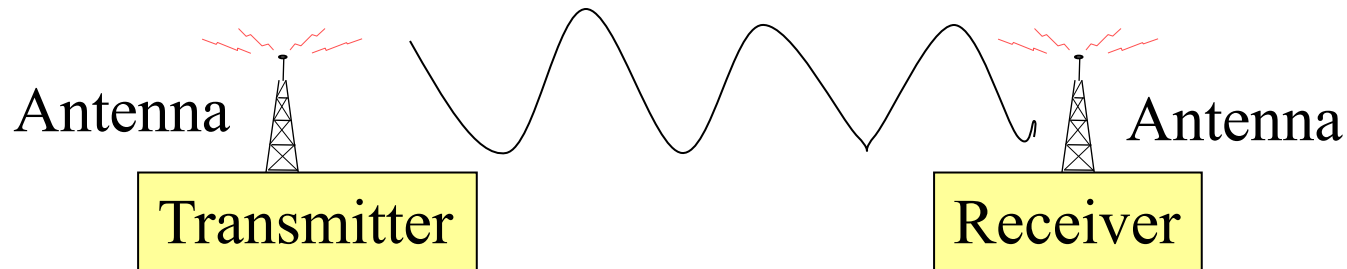
- Frequencies from 9KHz to 300 MHZ in high demand (especially VHF: 30-300MHZ)
- Two unlicensed bands
 - Industrial, Science, and Medicine (ISM): 2.4 GHz
 - Unlicensed National Information Infrastructure (UNII): 5.2 GHz
- Different agencies license and regulate
 - www.fcc.gov - US
 - www.etsi.org - Europe
 - www.wpc.dot.gov.in - India
 - www.itu.org - International co-ordination
- Regional, national, and international issues
- Procedures for military, emergency, air traffic control, etc

How does an antenna system work?

- Converts an electrical signal into electromagnetic radiation and vice versa
- Transmission
 - Radio transmitter supplies radio frequency current to antenna terminals
 - Antenna radiates energy from current in electromagnetic waves
- Reception
 - Antenna intercepts power from electromagnetic waves to produce voltage at its terminals, that voltage is then amplified

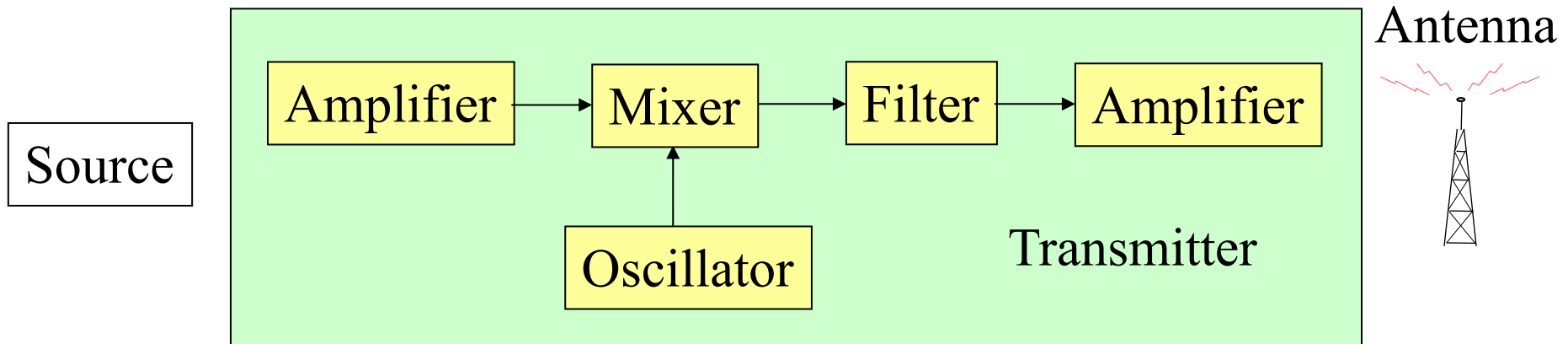


Wireless transmission



- Wireless communication systems consist of:
 - Transmitters
 - Antennas: radiates electromagnetic energy into air
 - Receivers
- In some cases, transmitters and receivers are on the same device, called transceivers.

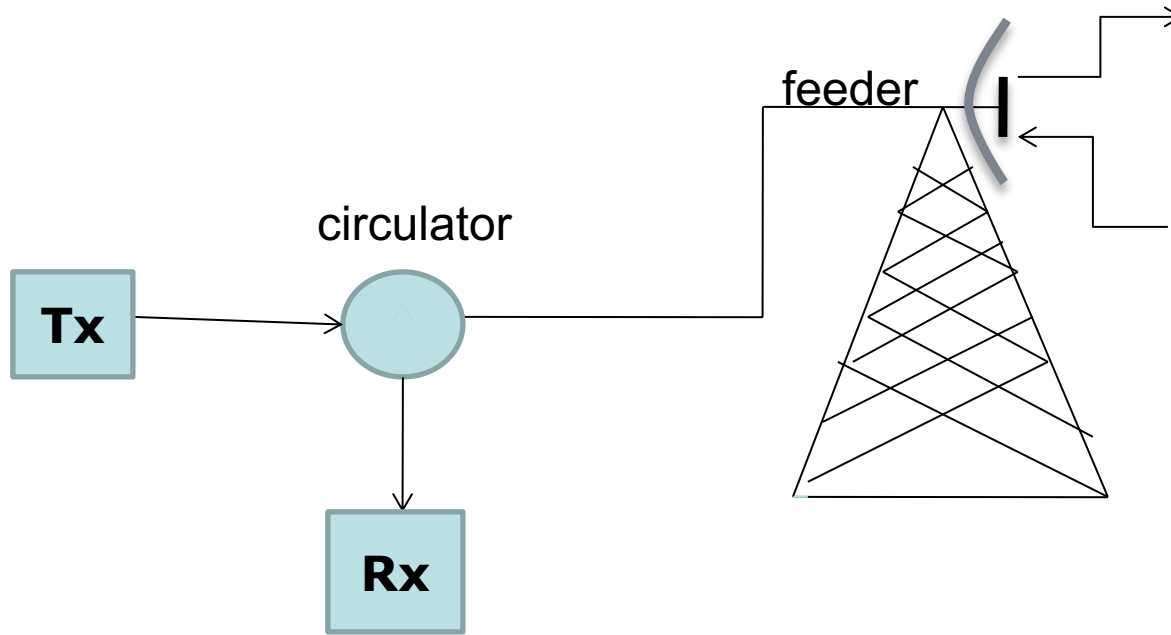
Transmitters



To generate a signal that is sent at 900 MHz and the original source generates a signal at 300 MHz you need

- (Pre-)Amplifier - strengthens the initial signal
- Oscillator - creates a carrier wave of 600 MHz
- Mixer - combines signal with oscillator and produces 900 MHz (also does modulation, etc.)
- Filter - selects correct frequency
- (Post-)Amplifier - Strengthens the signal before sending it

Antenna System Architecture



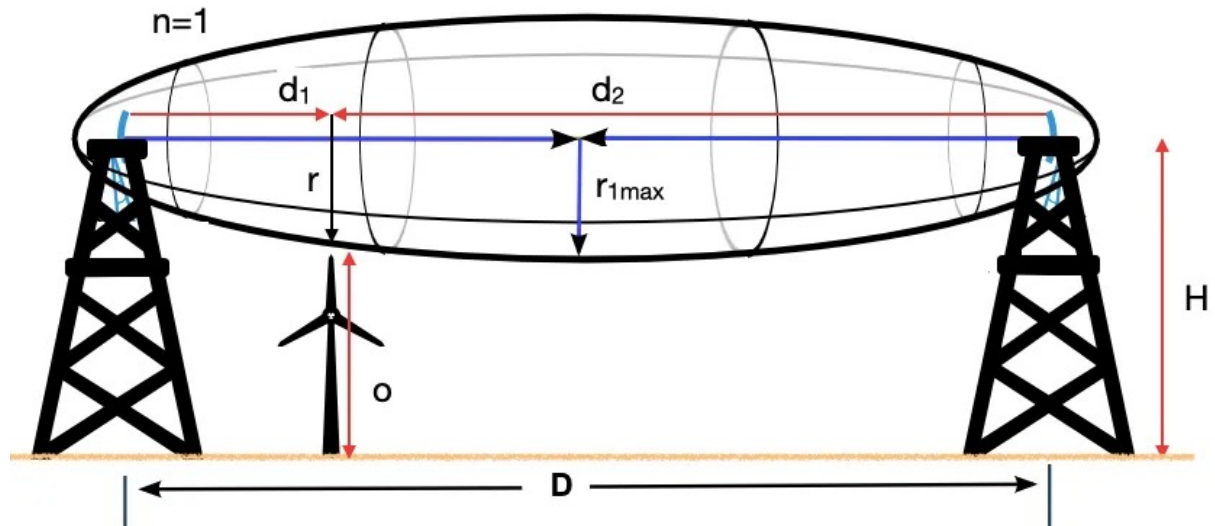
- **Radiating or receiving element:** is the physical mean transducing radio waves into electric current and vice versa
- **Feeder:** electric conductor media used for connecting transmitter and receiver with the radiating element
- **Circulator:** used for using the same antenna in both directions to separate transmission from reception

Features of a radio transmission systems

- The three main parameters that define a digital transmission system:
 - **Over-the-air data rate**
 - Determined by data throughput requirements
 - **Receive Sensitivity**
 - Indicates the level of signal strength that must be present to correctly receive data at a specified bit-error rate
 - **Transmit power**
 - It is usually driven by regulatory and power-consumptions considerations
 - For example, FCC allows up to 1 W of transmit power in the United States in the 2.4 Ghz band
- Transmission Range is a result of these three parameters and can be used to define them

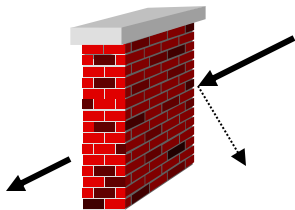
Propagation

- If unobstructed, radio waves will travel in a straight line from the transmitter to the receiver.
- But if there are obstacles near the path, waves reflecting off them may arrive out of phase with the signals that travel directly and reduce the power of the received signal.
- On the other hand, the reflection can enhance the power of the received signal if the reflection and the direct signals arrive in phase.
- The area in which an object near the path can influence the quality and power of signal is known as “Fresnel Zone”

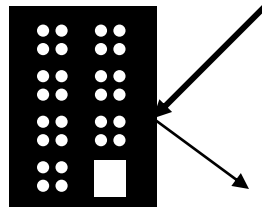


Propagation

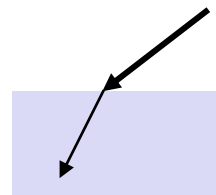
- Received power is proportional to $1/D^2$ (decays as the square of D = distance between sender and receiver)
- Received power is additionally influenced by
 - **fading** (frequency-dependent attenuation of a signal)
 - **shadowing** (power fluctuations due to obstacles)
 - **reflection** at large obstacles
 - **refraction** propagation drift depending on the varying density of a medium
 - **scattering** (deviation of reflected radiation from the angle predicted by reflection laws) at small obstacles
 - **diffraction** (waves bending around the corners of an obstacle) at edges



shadowing



reflection



refraction



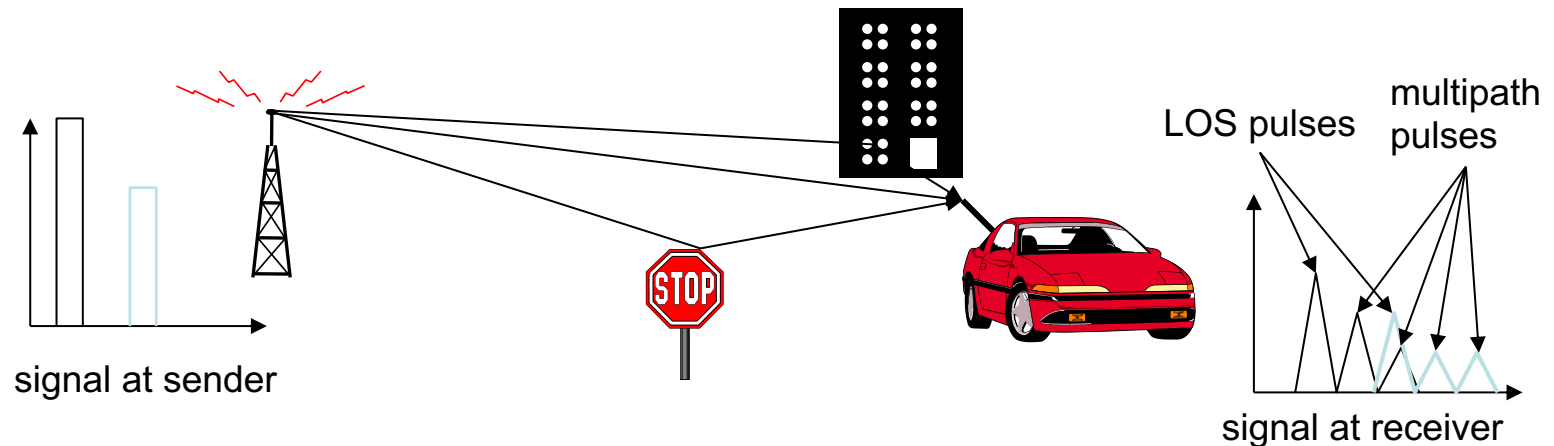
scattering



diffraction

Multipath propagation

- Signal can take many different paths between sender and receiver due to reflection, scattering, diffraction



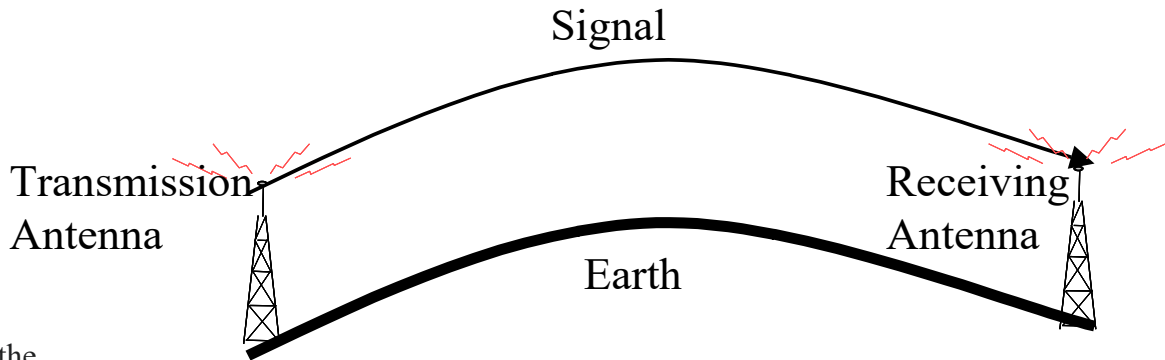
- Time dispersion: signal is dispersed over time
 - ➔ interference with "neighbor" symbols, Inter Symbol Interference (ISI)
- The signal reaches a receiver directly and phase shifted
 - ➔ distorted signal depending on the phases of the different parts

Propagation modes

For **low-frequency** waves

a) Ground Wave Propagation

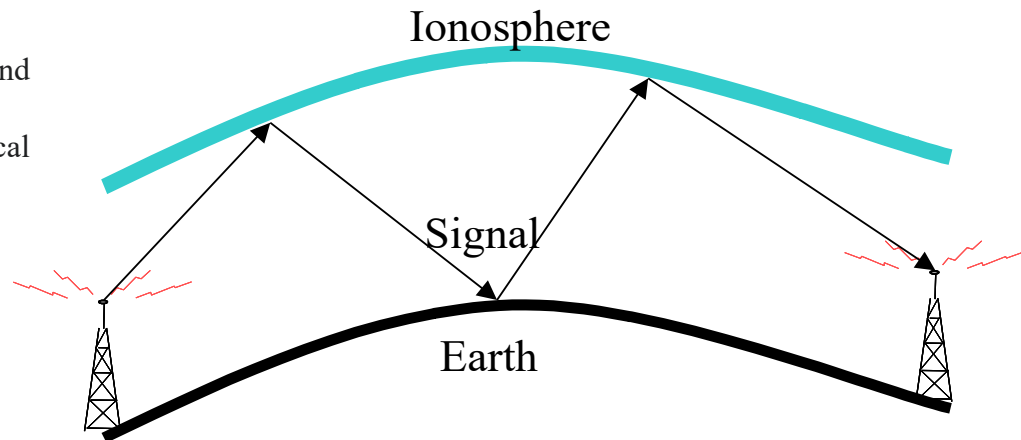
- Waves propagate by interacting with the conductive surface of the Earth following its curvature, by traveling over mountains and beyond the horizon.
- waves propagate in vertical polarization so vertical antennas (monopoles) are required.



For **long-distance high-frequency** waves

b) Sky Wave Propagation

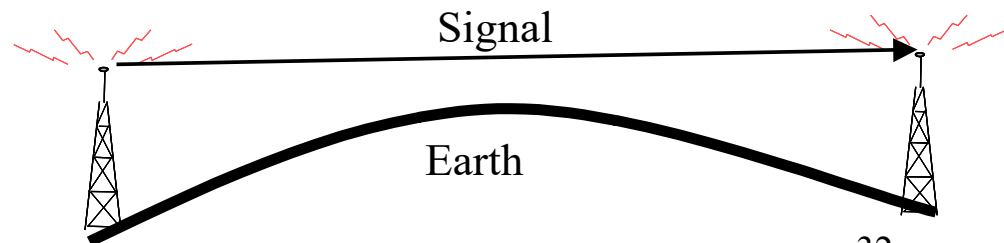
- Ionosphere contains layers of charged particles refracting radio waves back toward the Earth.
- A radio wave directed at an angle into the sky can be reflected back to Earth beyond the horizon by these layers, allowing long-distance radio transmission.



For **very high-frequency** waves

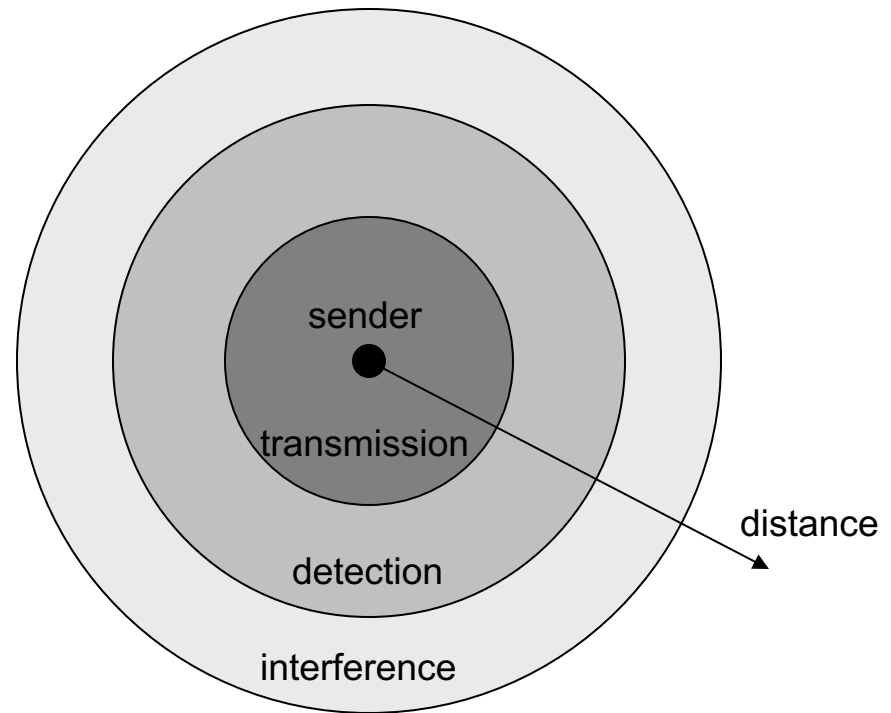
c) Line-of-Sight Propagation

- refers to radio waves that travel directly in a line from transmitter antenna to receiver one (max 64 Km)
- It does not necessarily require a cleared sight path; at lower frequencies waves can pass through obstructions



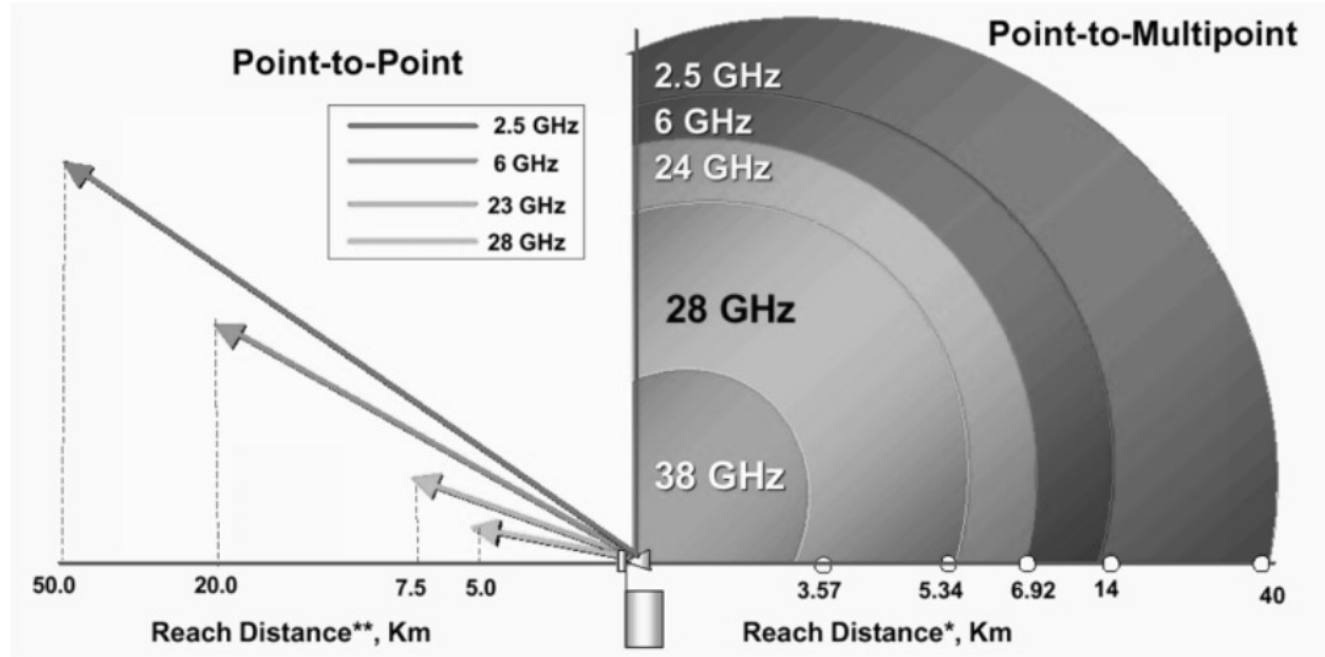
Propagation ranges

- Transmission range
 - communication possible
 - low error rate
- Detection range
 - detection of the signal possible
 - no communication possible
- Interference range
 - signal may not be detected
 - signal adds to the background noise



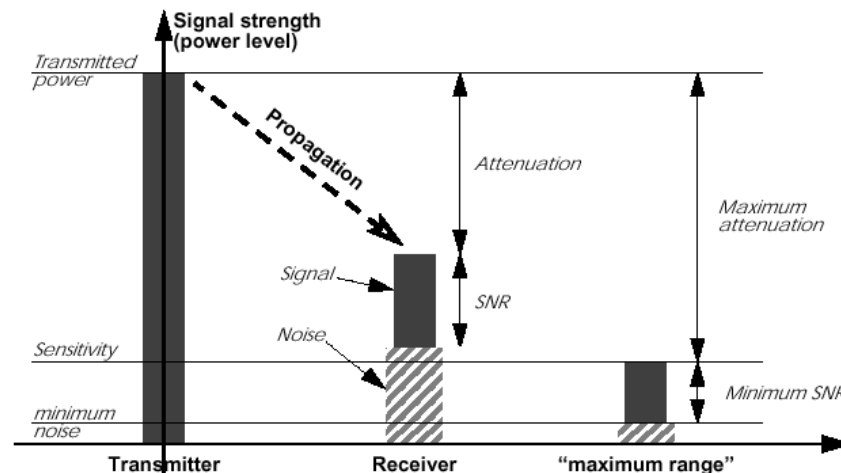
Range Depends On...

- Frequency
- Transmit power
- Radio sensitivity
- Processing gain from access technique and redundancy
- Interference effects



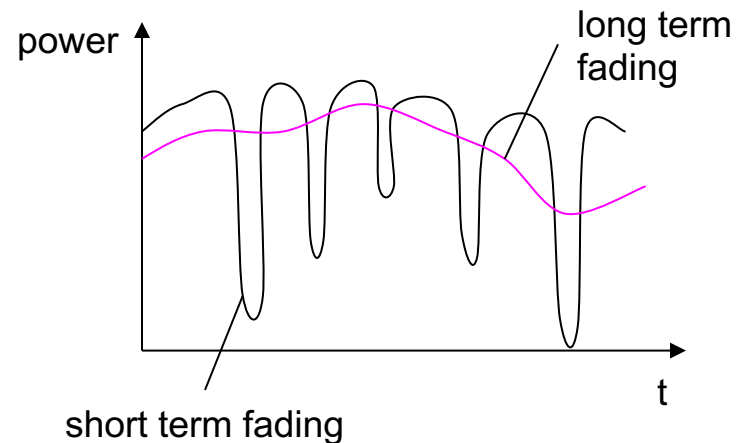
Attenuation

- Strength of signal falls off with distance over transmission medium
- Attenuation factors for unguided media:
 - Received signal must have sufficient strength so that circuitry in the receiver can interpret the signal
 - Signal must maintain a level sufficiently higher than noise to be received without error
 - Attenuation is greater at higher frequencies, causing distortion
- Approach: amplifiers that strengthen higher frequencies



Effects of mobility

- Channel characteristics change over time and location
 - signal paths change
 - different delay variations of different signal parts
 - different phases of signal parts
- ➔ quick changes in the power received (short term fading)

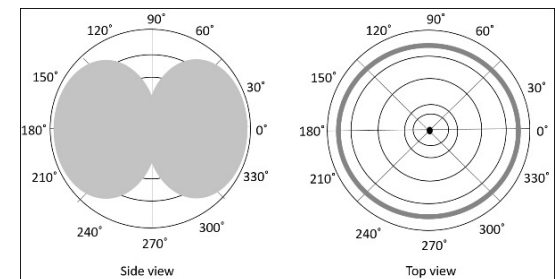
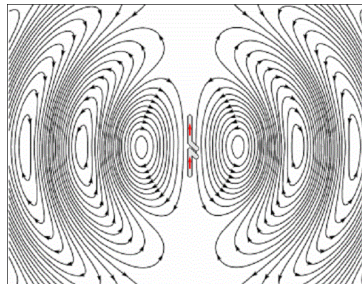
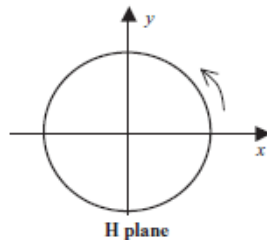
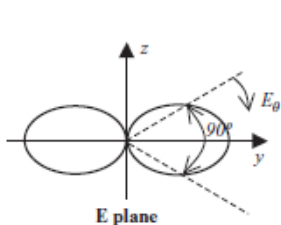


- Additional changes in
 - distance to sender
 - obstacles further away

- ➔ slow changes in avg power received (long term fading)

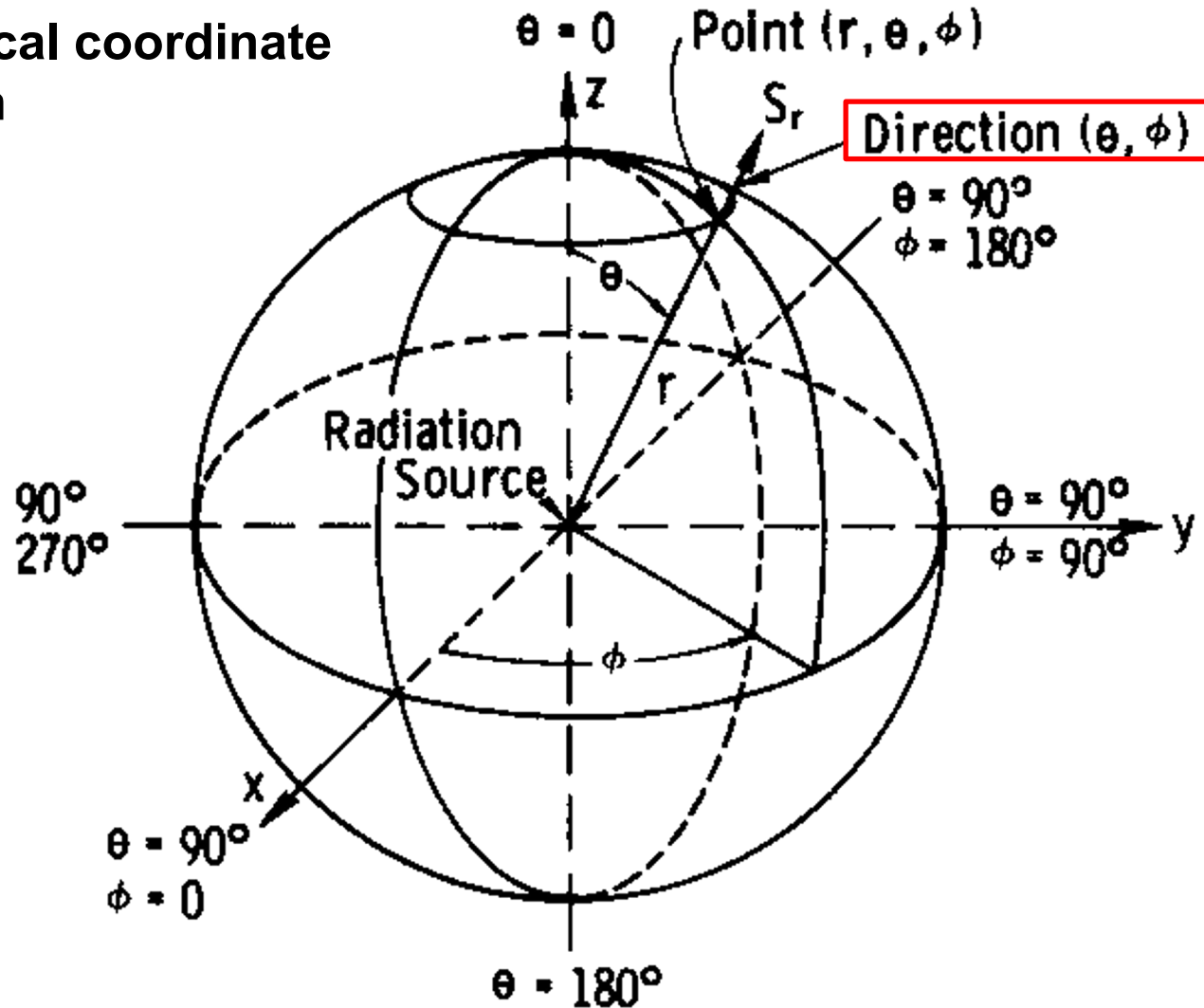
Antenna radiation

- The energy radiated by an antenna can be represented by the Radiation pattern of the antenna.
 - Radiation Patterns are diagrammatical representations of the distribution of radiated energy into space, as a function of direction
 - Graphically, radiation can be plotted as a function of angular position and radial distance from the antenna.
 - This is a mathematical function of radiation properties of the antenna represented as a function of spherical coordinates, $E(\theta, \phi)$ and $H(\theta, \phi)$
 - The E-plane is the plane that the electric field E lies on. These are the elevation planes
 - the H-plane is the plane that the magnetic field H is on, which is the azimuth plane.



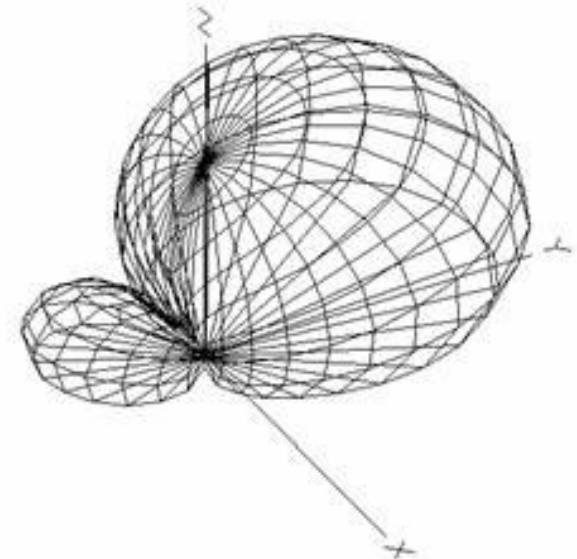
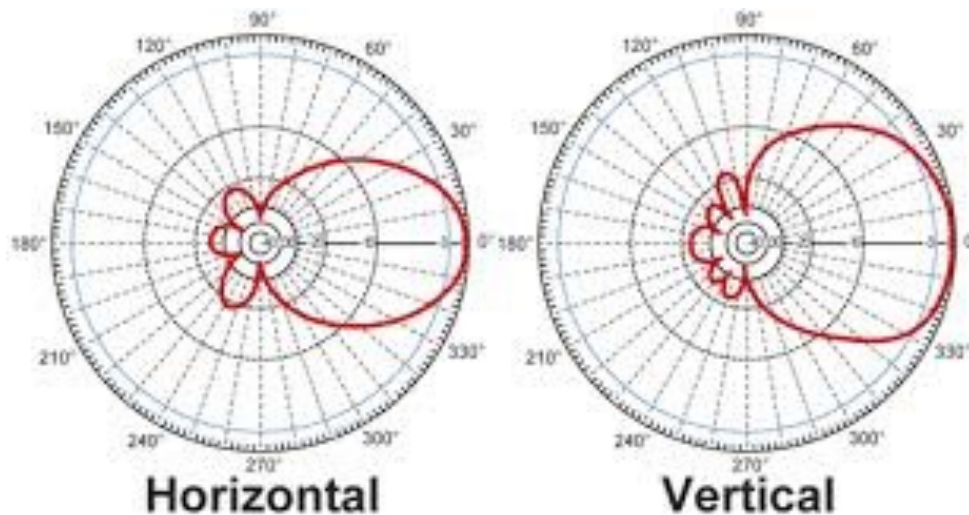
Antenna radiation

Spherical coordinate system



Antennas: radiation patterns

- The radiation pattern of an antenna is the variation of the field intensity of an antenna as an angular function with respect to the axis
- 2d radiation pattern
- 3d radiation pattern



- The radiation pattern of an antenna is the same regardless of whether it is used to transmit or receive a signal

Antennas: radiation patterns

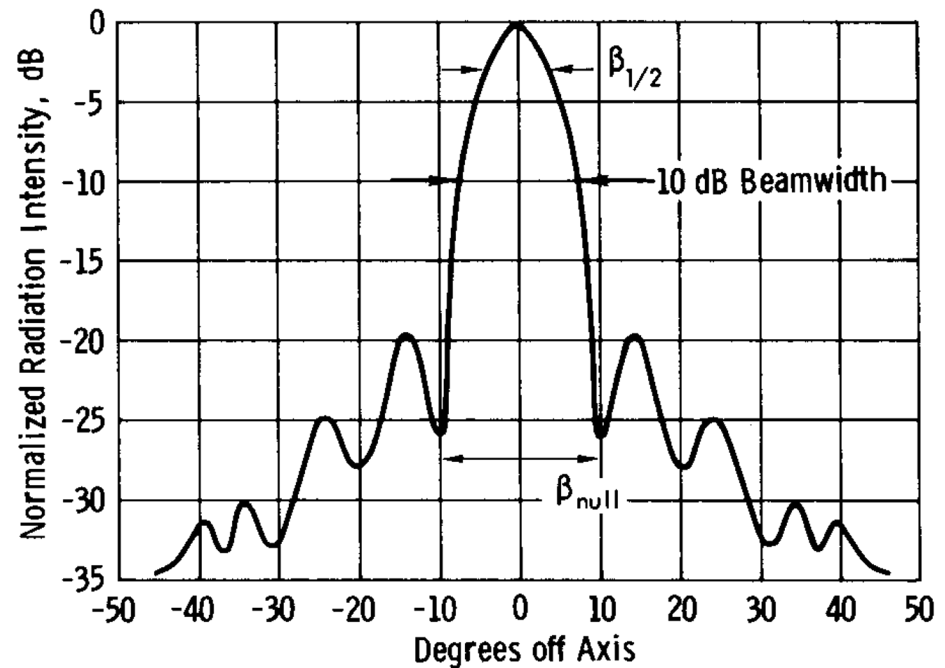
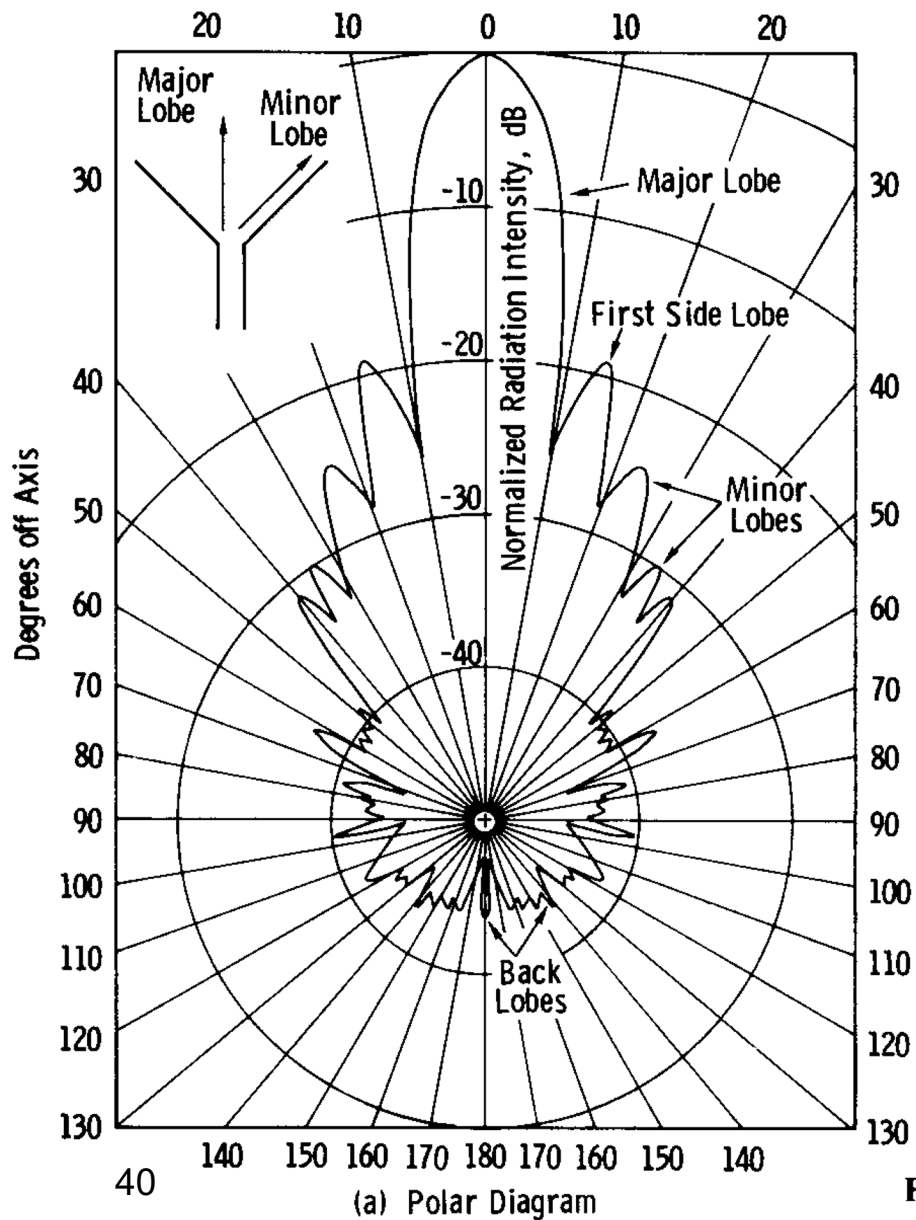
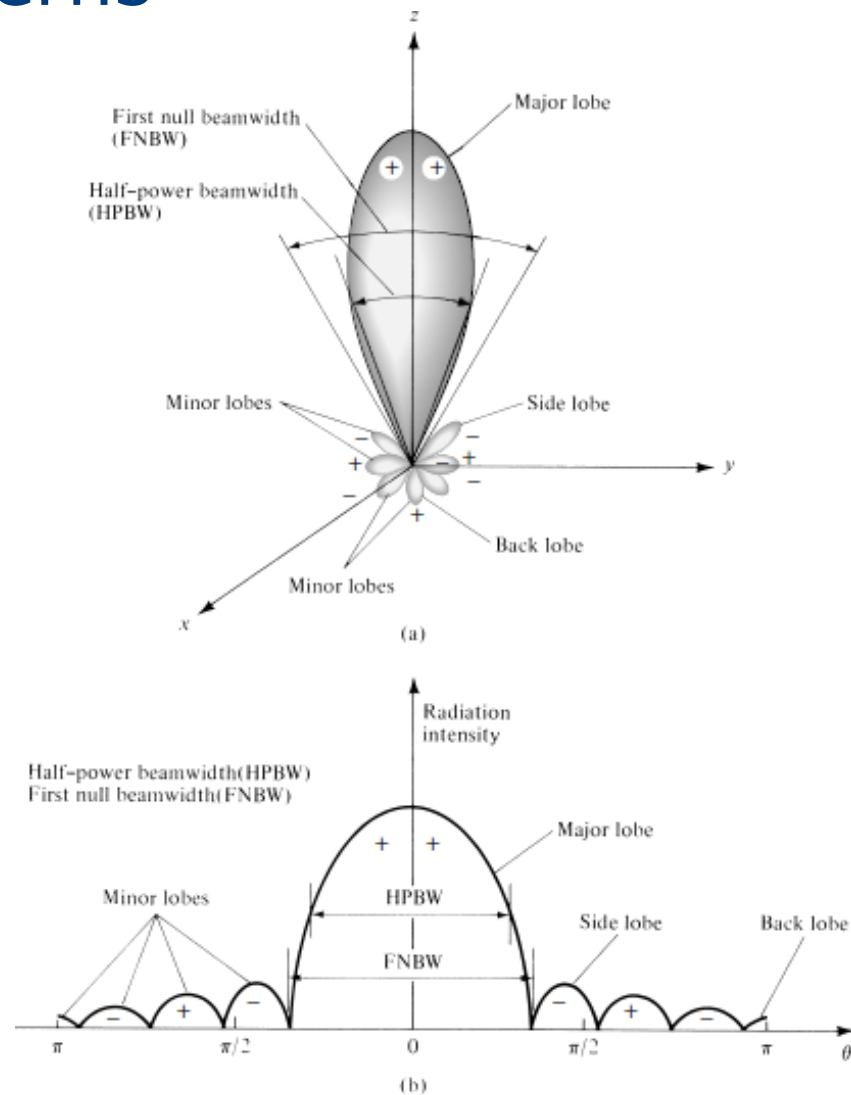


Fig. 3.3 Representative plots of the normalized radiation pattern of a microwave antenna in (a) polar form and (b) rectangular form.

Antennas: radiation patterns

- Various parts of a radiation pattern are referred to as lobes, ie. Main lobe, minor lobe, and back lobes
- The type of lobes depends on the radiation intensity at that part of the radiation pattern
- A major or main lobe is the radiation lobe containing the direction of maximum radiation (From the fig, $\theta = 0$ degrees)
- A minor lobe is any lobe other than the main lobe.
- A side lobe is usually adjacent to the main lobe
- A back lobe is the radiation lobe whose angle makes an angle of 180 degrees to the main lobe (opposite to the main lobe)



Antennas: isotropic radiator

- Isotropic antenna provides equal radiation in all directions (three dimensional)
- It is only a theoretical reference antenna
 - Real antennas always have directive effects (vertically and/or horizontally)
 - Since it is entirely non-directional, it serves as a hypothetical worst-case against which directional antennas may be compared
 - Having zero directivity it is used as a base of comparison to calculate the directivity of actual antennas

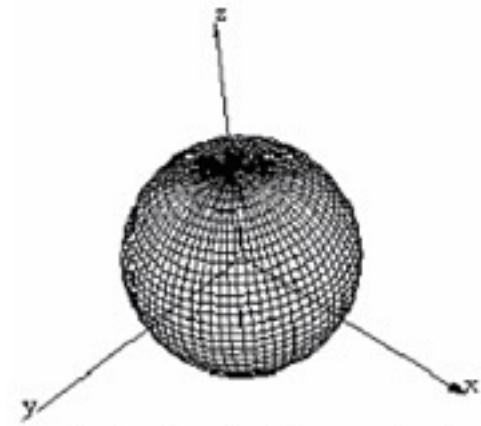
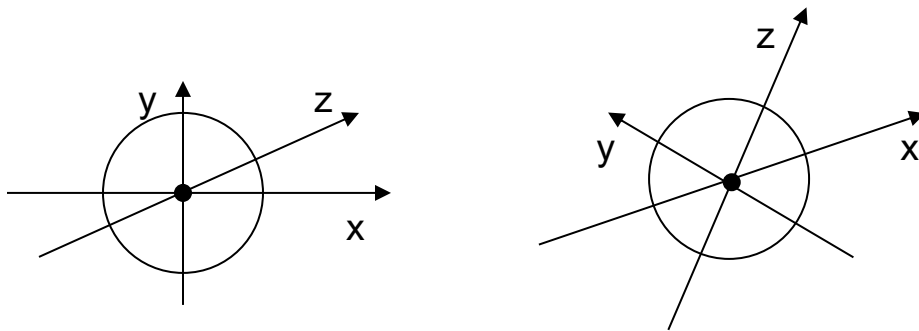


Fig.1. Isotropic Antenna Pattern

Antenna Selection

- Directivity
 - ability to focus in a particular direction to transmit or receive energy
 - Omni (360 degree coverage) directional
 - Directional (limited range of coverage)
- Gain
 - Relation between the power produced on the beam axis and the one produced by a hypothetical lossless isotropic antenna that equally sensitive to signals from all directions
 - combines the antenna's directivity and electrical efficiency
 - more gain means more coverage
- Bandwidth
 - is the range of operating frequencies
- Polarization
 - polarization of the radiated fields produced by an antenna (orientation of EM field vectors at a given point and how they vary with time)

Directivity

Formally is the ratio of the radiation intensity in a given direction from the antenna to the radiation intensity averaged over all directions.

$$D(\theta, \phi) = \frac{F_n(\theta, \phi)}{\frac{1}{4\pi} \iint_{4\pi} F_n(\theta, \phi) d\Omega} \quad [\text{unitless}]$$

Maximum directivity, D_o , found in the direction (θ, ϕ) where $F_n = 1$

$$D_0 = \frac{4\pi}{\iint_{4\pi} F_n(\theta, \phi) d\Omega} = \frac{4\pi}{\Omega_p} \quad \text{and} \quad \Omega_p \simeq \beta_{xz}\beta_{yz} \quad \text{or} \quad D_0 = \frac{4\pi}{\Omega_p} \simeq \frac{4\pi}{\beta_{xz}\beta_{yz}}$$

Given D_o , D can be found

$$D(\theta, \phi) = D_0 F_n(\theta, \phi)$$

Gain

Gain – ratio of the power at the input of a loss-free isotropic antenna to the power supplied to the input of the given antenna to produce, in a given direction, the same field strength at the same distance

Of the total power P_t supplied to the antenna, a part P_o is radiated out into space and the remainder P_l is dissipated as heat in the antenna structure. The *radiation efficiency*, defined as the ratio of P_o to P_t is:

$$\eta_l = \frac{P_o}{P_t}$$

Therefore gain, G , is related to directivity, D , as

$$G(\theta, \phi) = \eta_l D(\theta, \phi)$$

And maximum gain, G_o , is related to maximum directivity, D_o , as

$$G_o = \eta_l D_o$$

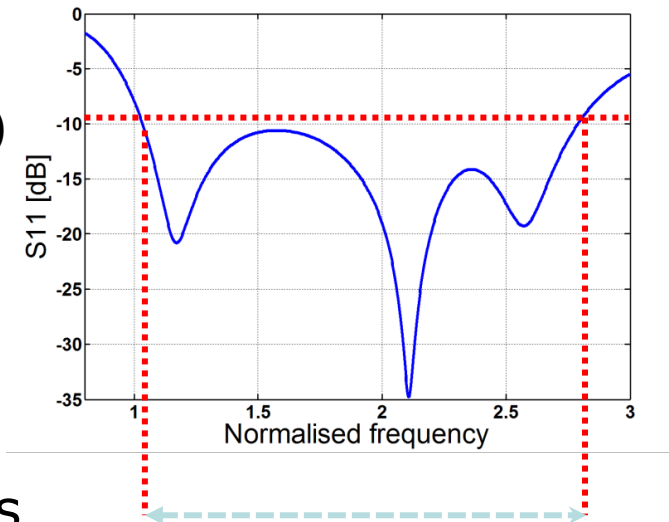
Bandwidth

The antenna's bandwidth is the range of operating frequencies over which the antenna can properly radiate and receive energy by meeting its operational requirements such as:

- Spatial properties (radiation features)
- Polarization properties
- Impedance properties
- Propagation mode properties

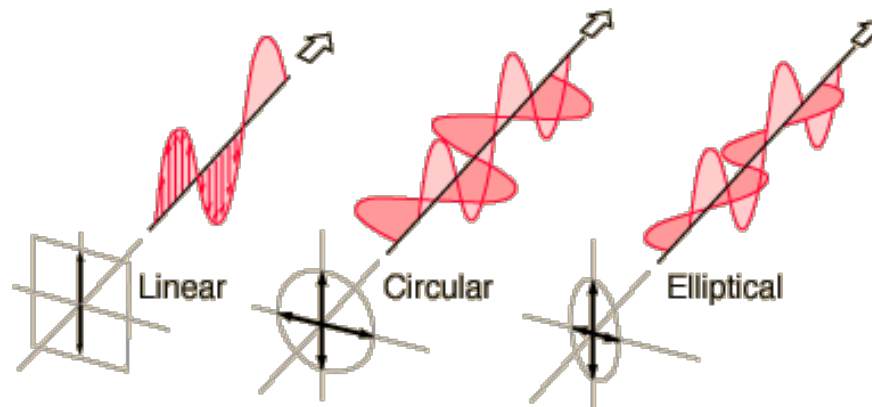
Most antenna technologies can support operation over a frequency range that is 5 to 10% of the central frequency

To achieve wideband operation requires specialized technologies



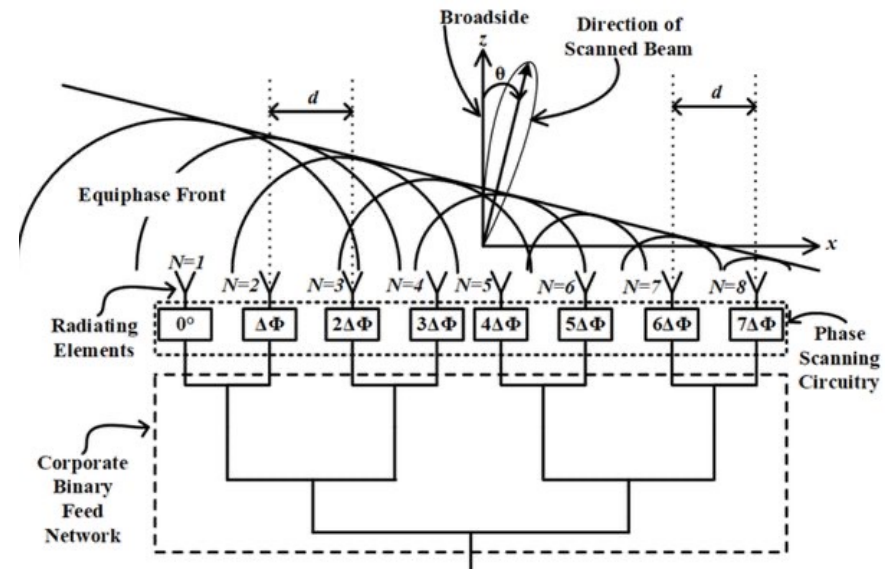
Polarization

- Antenna Polarization is defined at the orientation of the electric fields produced and radiated by the antenna.
- Antennas have the property of being more sensitive **to one polarization** than another which provides the ability to filter signals based on its polarization
- The type of polarization generated by the antenna depends on how the currents move in/on the antenna
 - For **Linear polarization**, the currents should travel along one axis
 - For **circular** or **elliptical** polarization, the antenna produces two orthogonal currents with equal magnitude and with 90 degrees phase shift between the currents.



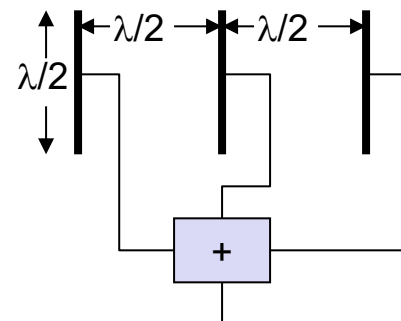
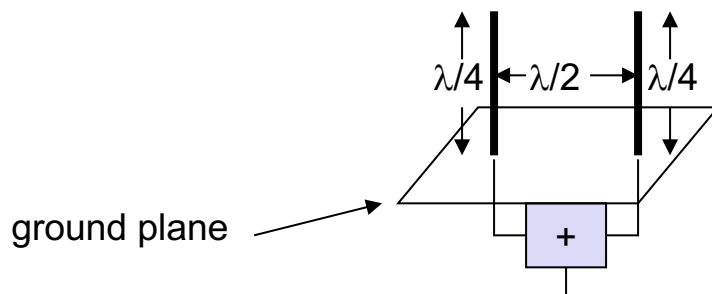
Spatial filtering

- Antennas have the property of being more sensitive in **one direction** than in another which provides the ability to spatially filter signals from its environment.
- Spatial filtering is a signal processing technique for directional signal transmission or reception
 - This is achieved by combining elements in an antenna array in such a way that signals at particular angles experience constructive interference while others experience destructive interference.
 - It can be used at both the transmitting and receiving ends in order to achieve spatial selectivity.
 - A **beamformer** controls the phase and relative amplitude of the signal at each transmitter, in order to create a pattern of constructive and destructive interference in the wavefront.
 - When receiving, information from different sensors is combined in a way where the expected pattern of radiation is preferentially observed.



Diversity

- Grouping 2 or more antennas to improve link quality and reliability
- Multiple antennas offer several observations of the same signal
 - Each antenna will experience a different interference environment
 - Collectively such a system can provide a robust link
- Spatial diversity
 - multiple separated antennas, usually with the same characteristics
 - beneficial for mobile communication: allows multiple users to share a limited communication spectrum and avoid co-channel interference.
- Pattern diversity
 - two or more co-located antennas with different radiation patterns.
 - collectively capable of discriminating a large portion of angle space and can provide a higher gain versus a single omnidirectional radiator.
- Polarization diversity
 - pairing complementary polarizations, can immunize a system from polarization mismatches that would otherwise cause signal fade



Antenna Aperture

The antenna aperture, effective area, or receiving cross section, is a measure of how effective an antenna is at receiving the power of electromagnetic radiation

If an antenna delivers P_t watts to the load connected to its output terminals (e.g. the receiver) when irradiated by a uniform field of receiver power density S_r watts per square meter, the antenna's aperture A_r in square meters is given by:

$$A_r = P_t / S_r$$

A general relation for the effective aperture in terms of the peak antenna gain (G) of any antenna is given by:

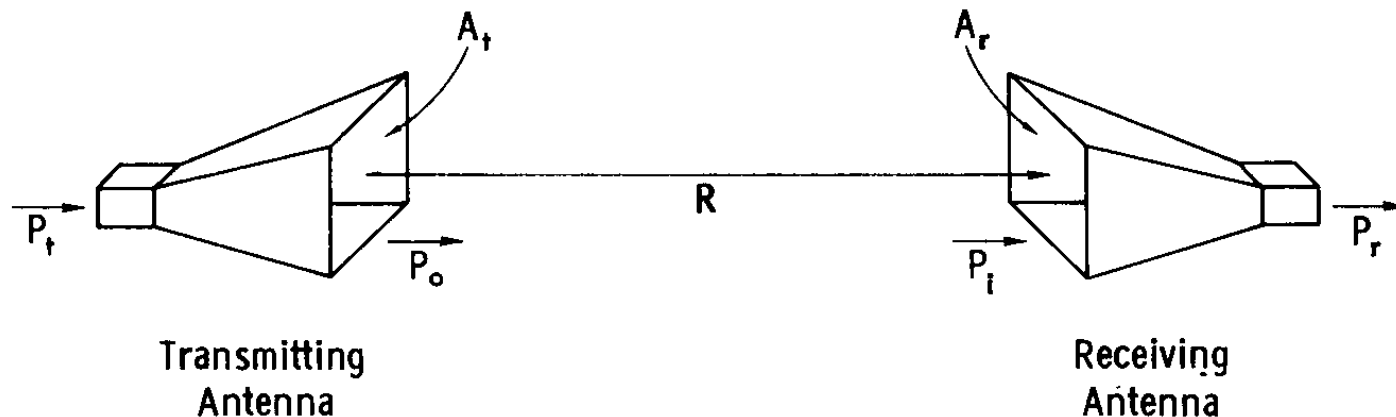
$$A_e = \frac{\lambda^2}{4\pi} G$$

Friis' transmission formula

At a fixed distance R from the transmitting antenna, the power intercepted by the receiving antenna with effective aperture A_r is

$$P_i = S_r A_r = \frac{P_t}{4 \pi R^2} G_t A_r$$

where S_r is the received power density (W/m^2), and G_t is the peak gain of the transmitting antenna.



Friis' transmission formula

If the radiation efficiency of the receiving antenna is η_r , then the power received at the receiving antenna's output terminals is

$$P_r = \eta_r P_i = \frac{P_t}{4 \pi R^2} G_t \eta_r A_r$$

Therefore we can write

$$\frac{P_r}{P_t} = \left(\frac{\lambda}{4 \pi R} \right)^2 G_t G_r = \eta_t \eta_r \frac{A_t A_r}{\lambda^2 R^2}$$

which is known as *Friis' transmission formula*

Friis' transmission formula

as Friis' transmission formula can be rewritten to explicitly represent the *free-space transmission loss*, L_{FS}

$$L_{FS} = \left(\frac{4 \pi R}{\lambda} \right)^2$$

which represents the propagation loss experienced in transmission between two lossless isotropic antennas.

With this definition, the Friis formula becomes

$$\frac{P_r}{P_t} = \frac{G_t G_r}{L_{FS}}$$

Friis' transmission formula

Finally, a general form of the Friis' transmission formula can be written that does not assume the antennas are oriented to achieve maximum power transfer

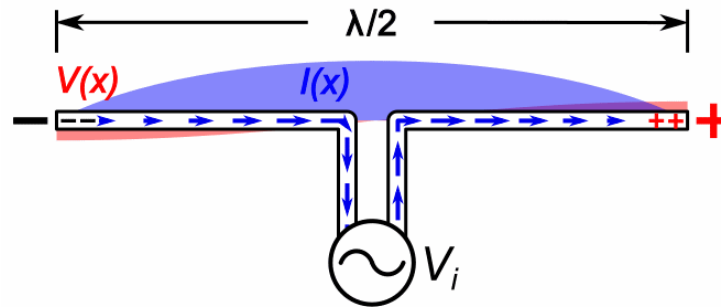
$$\frac{P_r}{P_t} = \left(\frac{\lambda}{4\pi R} \right)^2 G_t(\theta_t, \phi_t) G_r(\theta_r, \phi_r)$$

where (θ_t, ϕ_t) is the direction of the **receiving** antenna in the **transmitting** antenna coordinates, and vice versa for (θ_r, ϕ_r)

An additional term could be included to represent a polarization mismatch between the transmit and receive antennas.

Resonant and non-resonant antennas

- **Resonant antennas** are opened out transmission line i.e. they are open circuited at one end
- They have resonant lengths i.e. multiple of half-wave length.



- In **Non-resonant antennas** the source is matched to the load (i.e. they don't have open circuit).
- A non-resonant antenna is like a properly terminated transmission line, it produces no standing waves.
- They are suppressed by the use of a correct termination resistor and no power is reflected, ensuring that only forward travelling waves will present.

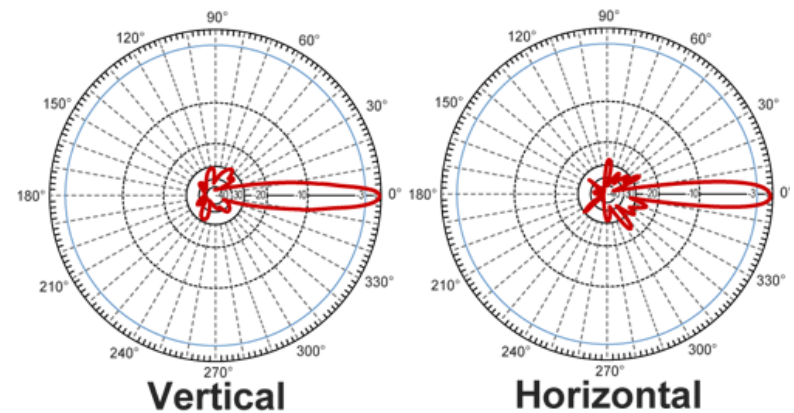
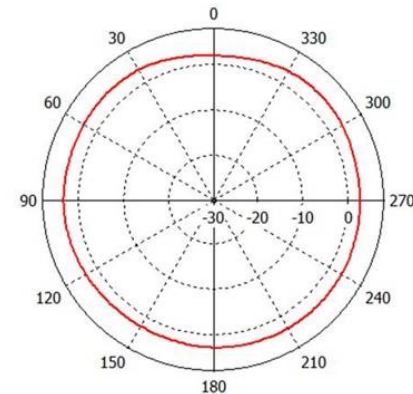
Antenna modes

- In **Omnidirectional** Mode:

- receive signals with gain G_o

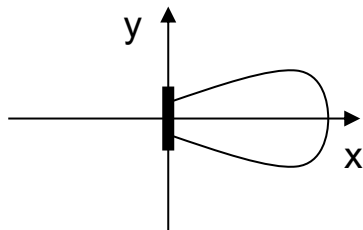
- In **Directional** Mode:

- Capable of beamforming in specified direction
 - Increase “range”
 - Gain is higher:
 - Directional Gain G_d ($G_d > G_o$)

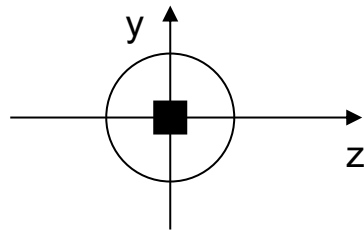


Directed and sectorized antennas

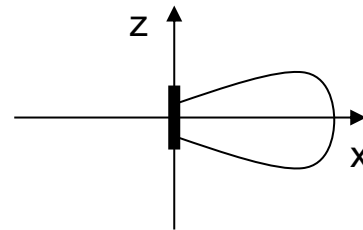
- A directed antenna radiates or receives greater power in specific directions allowing increased performance and reduced interference
- A sectorized antenna is a type of directional microwave antenna with a sector-shaped radiation pattern.
 - Often used as base stations for mobile devices (e.g., radio coverage of a valley)
 - To increase or widen the coverage area, several sector antennas are installed on the same supporting structure, e.g. tower



side view (xy-plane)

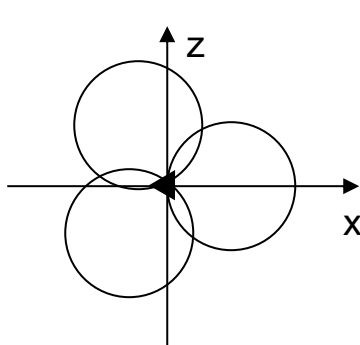


side view (yz-plane)

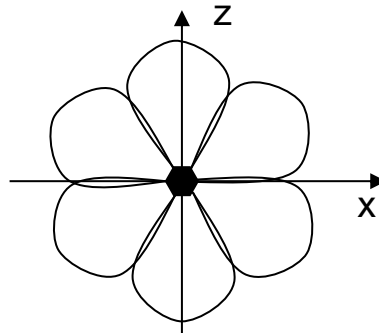


top view (xz-plane)

directed
antenna



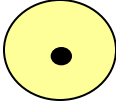
top view, 3 sector



top view, 6 sector

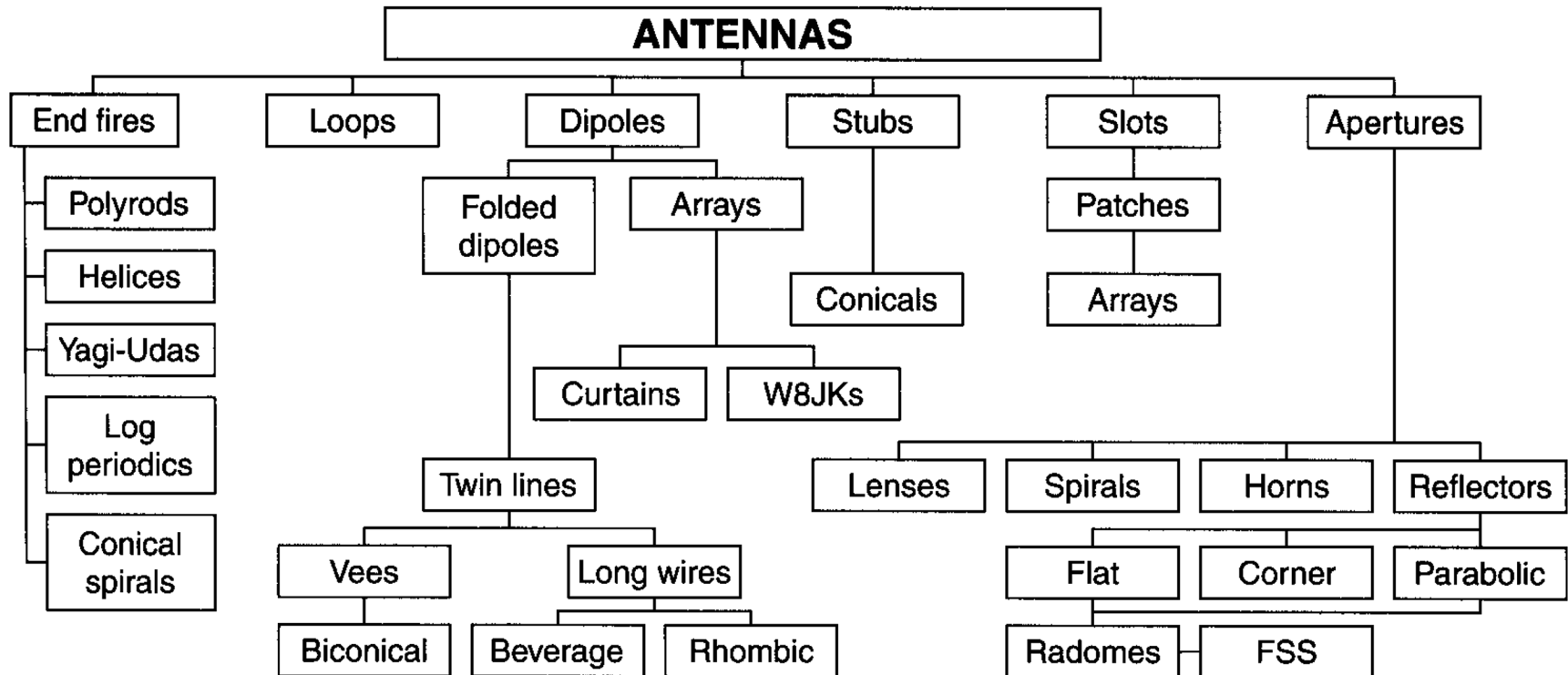
sectorized
antenna

Comparison of omni and directional

Issues	Omni	Directional
Spatial Reuse	Low	High
Connectivity	Low	High
Interference	Omni 	Directional 
Cost & Complexity	Low	High

Antenna types

- Antennas come in a wide variety of sizes and shapes



Antenna types

- Two types of antennas: **wire and aperture**
- Wire types include dipoles, monopoles, loops, rods, stubs, helices, Yagi-Uda, spirals.
- Aperture types include horns, reflectors, parabolic, lenses.

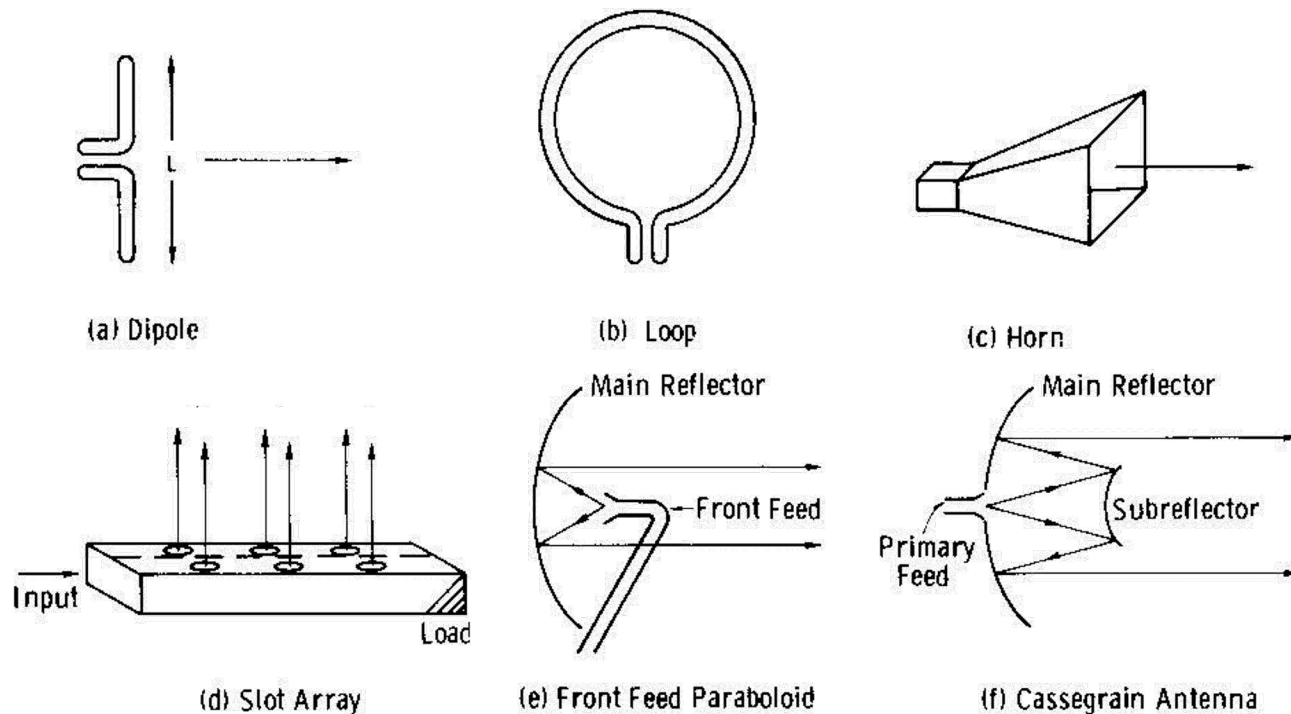
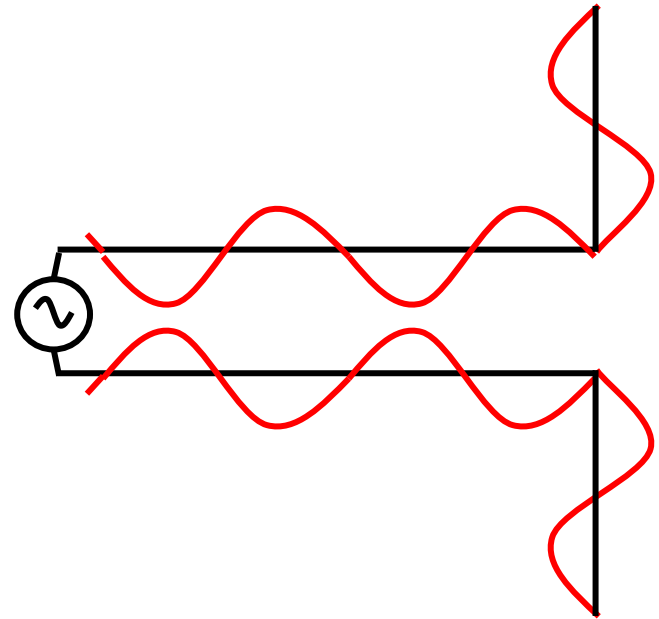


Fig. 3.1 Antenna types commonly used in microwave remote sensing.

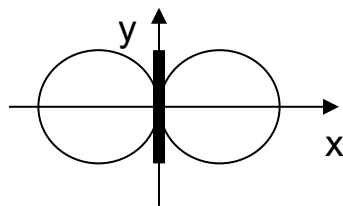
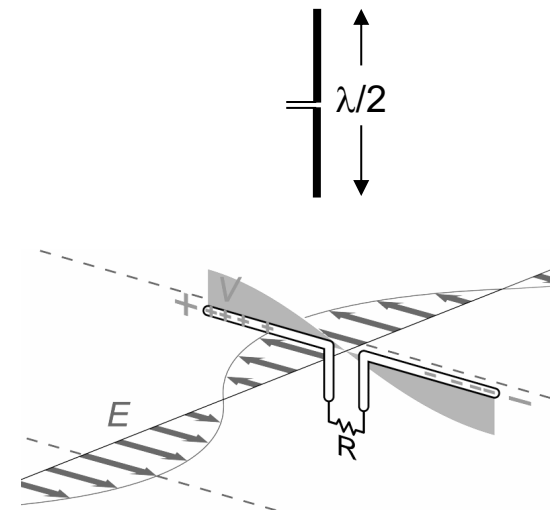
Wire Antennas

- In wire-type antennas the radiation characteristics are determined by the current wire distribution which produces the local magnetic field.
- By definition, the category of wire antennas includes all antennas formed of a conductor structure where, due to small diameter of cables, we consider only the linear current densities.
- The basic antennas are: dipoles, monopoles, loops.
- More advanced structures: helical, Yagis, the log-periodic ...

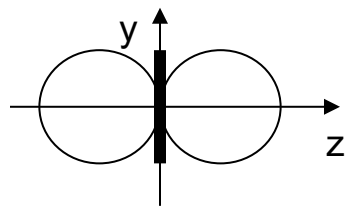


Wire antennas: simple dipoles

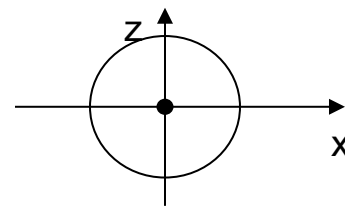
- Is the simplest antenna consisting of two conductors of equal length oriented end-to-end with the feedline connected between them
- Dipoles are frequently used as resonant antennas.
 - able to resonate at a particular frequency, just like a plucked guitar string
 - using the antenna at around that frequency is advantageous in terms of feedpoint impedance so its length is determined by the intended wavelength (or frequency) of operation (the shape of antenna proportional to wavelength λ)
 - dipoles with length $\lambda/2$ are known as Hertzian dipoles
 - the electric field (E) of the incoming wave pushes the electrons in the conductors back and forth, charging the ends alternately positive (+) and negative (–)
 - since the length of the antenna is one half the wavelength, the oscillating field induces standing waves of voltage (V) and current in the conductors
 - the oscillating currents flow down the transmission line and through the receiver (the resistance R).



side view (xy-plane)



side view (yz-plane)

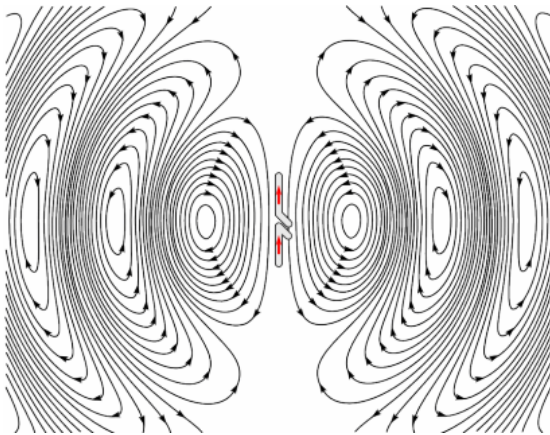


top view (xz-plane)

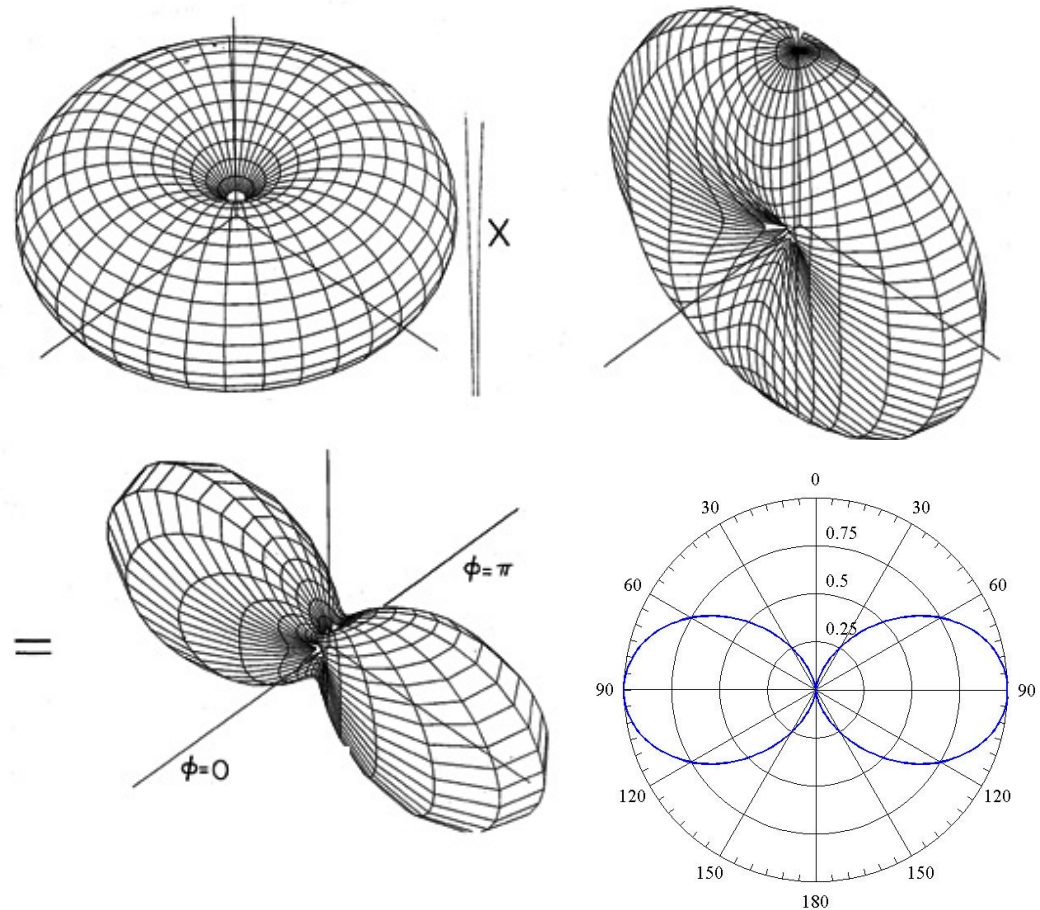
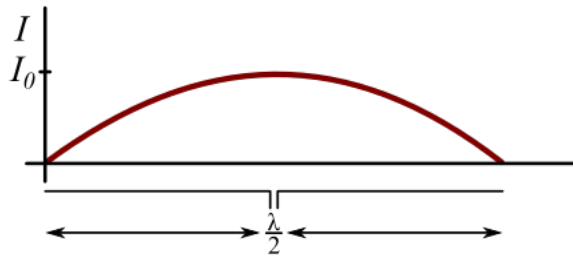
simple
dipole

Dipole antenna

Achieves maximum power in the direction of the main lobe compared an isotropic radiator (with the same average power)



Electric field of a half-wave dipole (tx)

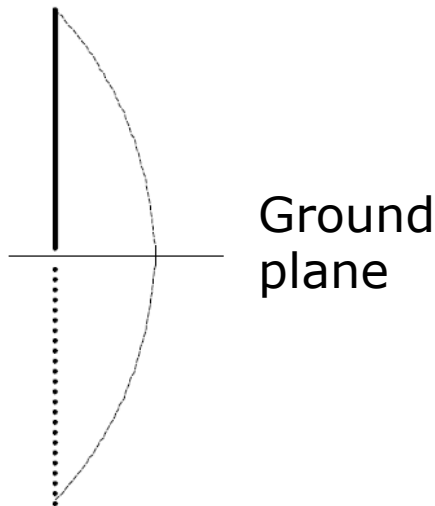


The magnitude of current in a standing wave along the dipole.

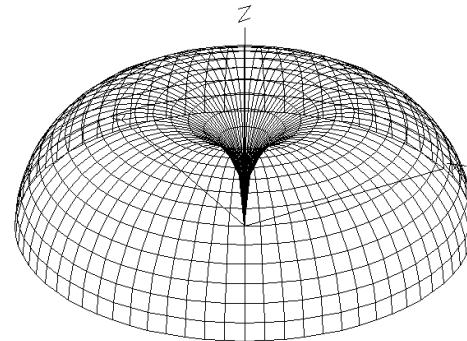
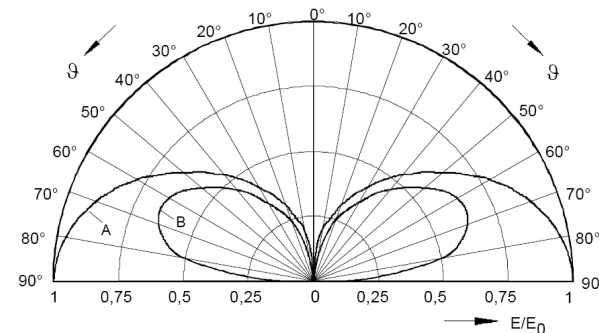
Radiation pattern of dipole antenna

Wire antennas: monopole antenna

- The monopole is a straight rod-shaped conductor mounted perpendicularly over some type of conductive surface, the ground plane
- The signal is applied (tx), or taken (rx), between the lower end of the monopole and the ground plane in contrast with dipole where the signal is applied between the two halves of the antenna.
- Operate at lower frequencies, $<20\text{MHz}$, with the earth as ground plane



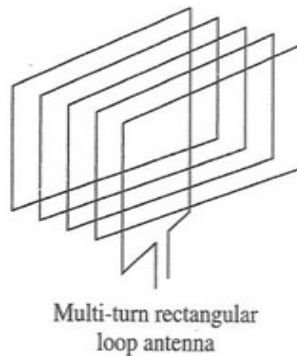
Mirroring principle creates image of monopole, transforming it into a dipole



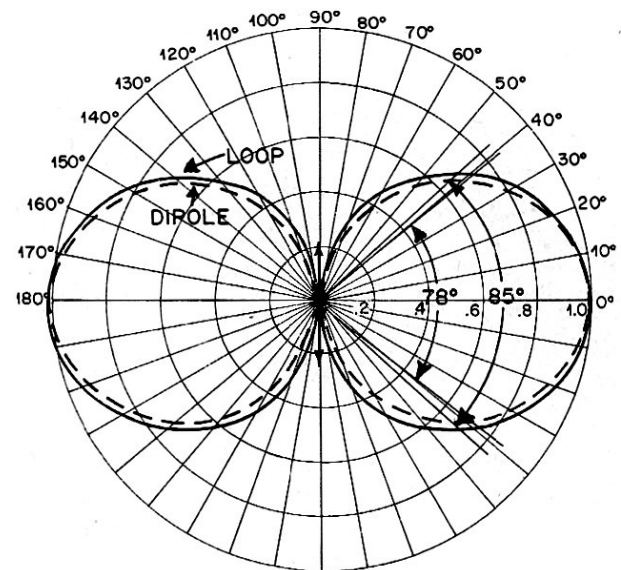
Radiation pattern of vertical monopole above ground

Wire antennas: loops

- consist of a loop of wire or other electrical conductor, that is usually fed by a balanced source. There are two distinct types:
 - **Large loop** antennas (or self-resonant loop antennas):
 - have a perimeter close to one or more whole wavelengths at the operating frequency, which makes them self-resonant at that frequency
 - are the most efficient of all antenna types for both transmission and reception.
 - **Small loop** antennas (or magnetic loops)
 - have a perimeter smaller than half the operating wavelength (no more than $1/4$)
 - are used mainly as receiving antennas, but are sometimes used for transmission

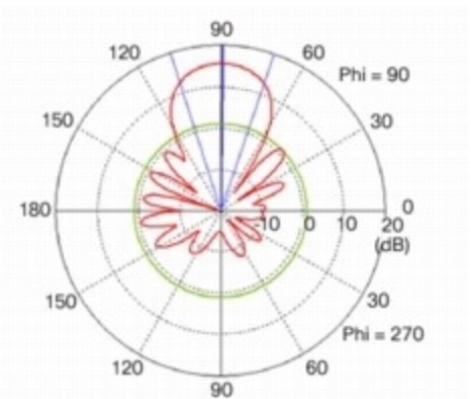
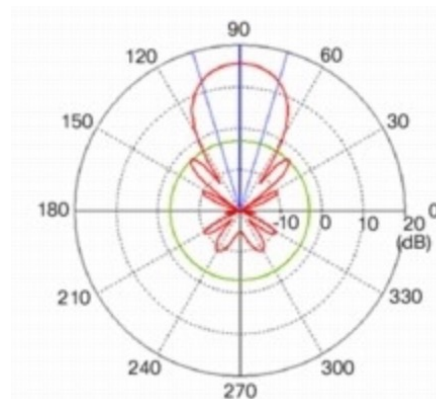
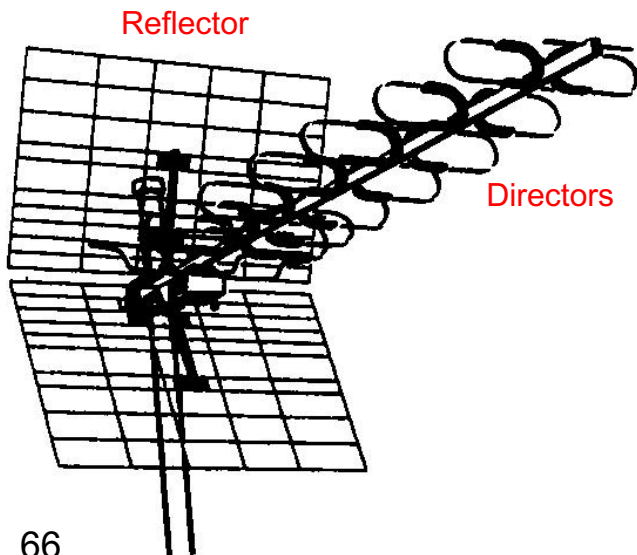


Large loop antennas have a two-lobe radiation pattern at their first, full-wave resonance, peaking in both directions perpendicular to the plane of the loop.



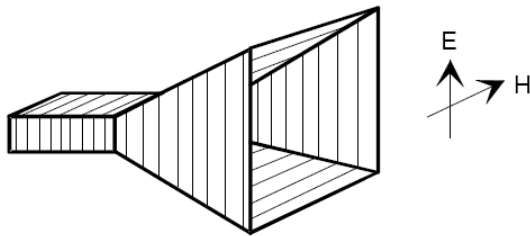
Wire antennas: Yagi-Uda

- A Yagi-Uda is a directional antenna consisting of a long transmission line with a single driven element consisting of:
 - two rods connected on either side of the transmission line.
 - a single reflector on one side of the transmission line, a folded dipole and a number of parasitic elements which act as directors
 - The more directors, the greater the gain and the longer is the antenna
- It propagates EM field energy in the direction from the driven element toward the director(s), and is most sensitive to incoming EM field energy in this same direction

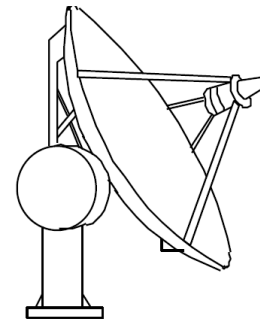


Aperture antennas

- In **aperture-type antennas** EM waves are emitted through an opening (or aperture)
- the radiation characteristics are determined by the field distribution across the aperture
 - close analogs in acoustics: megaphone and parabolic microphone.
- They are used at microwave frequencies to obtain narrow beams and achieve high gain
- Examples: waveguides, horns, Parabolic reflectors and patches



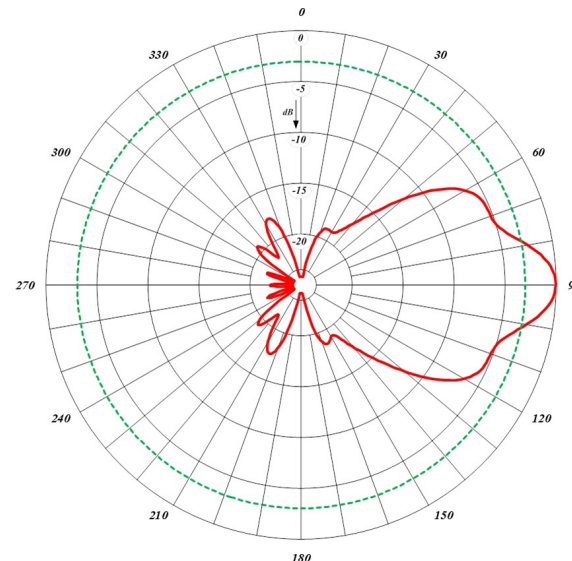
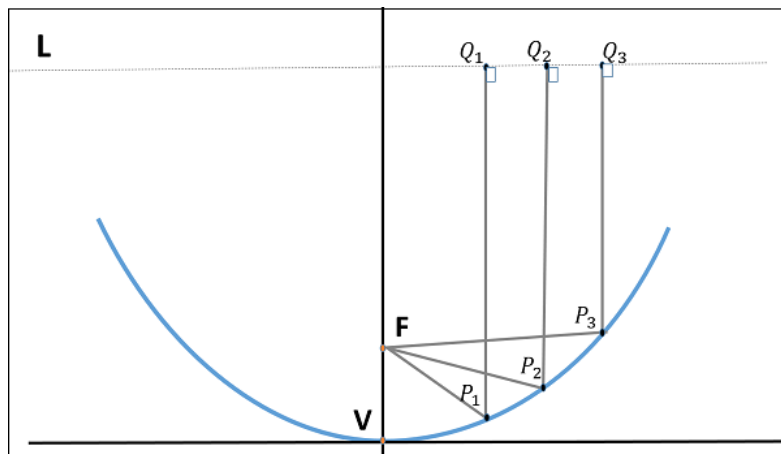
Horn antenna



Parabolic reflector

Aperture antennas: parabolic reflector

- Uses a parabolic reflector, a curved surface with the cross-sectional shape of a parabola, to direct the radio waves
 - In transmission the signal from the feed, comes out of a dipole or a horn antenna, to focus the wave on the parabola
 - the waves come out of the focal point (F) and strike the reflector
 - the wave gets reflected as a collimated wave front, to be transmitted
 - When receiving, EM wave hits the shape of the parabola
 - it gets reflected onto the feed point.
 - The dipole or the horn antenna, which acts as the receiver antenna at its feed, receives this signal, converts it into electric and forwards it to the receiver circuitry
- The main advantage is high directivity
- used as high-gain antennas for point-to-point communications



Aperture antennas: patch

- consists of a planar sheet or "patch" of metal, mounted over a larger sheet of metal called a ground plane
 - the two metal sheets together form a resonant piece of microstrip transmission line with a length of approximately one-half wavelength of the radio waves.
 - The radiation mechanism arises from fringing fields along the radiating edges
 - The radiation at the edges causes the antenna to act slightly larger electrically than its physical dimensions, so in order for the antenna to be resonant, a length of microstrip transmission line slightly shorter than one-half the wavelength of frequency used.
 - It is mainly practical at microwave frequencies, at which wavelengths are short enough that the patches are conveniently small and is widely used in portable wireless devices because of the ease of fabricating it on printed circuit boards

